



# Federated learning-guided intrusion detection and neural key exchange for safeguarding patient data on the internet of medical things

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## Abstract

To improve the security of the Internet of Medical Things (IoMT) in healthcare, this paper offers a Federated Learning (FL)-guided Intrusion Detection System (IDS) and an Artificial Neural Network (ANN)-based key exchange mechanism inside a blockchain framework. The IDS are essential for spotting network anomalies and taking preventative action to guarantee the secure and dependable functioning of IoMT systems. The suggested method integrates FL-IDS with a blockchain-based ANN-based key exchange mechanism, providing several important benefits: (1) FL-based IDS creates a shared ledger that aggregates nearby weights and transmits historical weights that have been averaged, lowering computing effort, eliminating poisoning attacks, and improving data visibility and integrity throughout the shared database. (2) The system uses edge-based detection techniques to protect the cloud in the case of a security breach, enabling quicker threat recognition with less computational and processing resource usage. FL's effectiveness with fewer data samples plays a part in this benefit. (3) The bidirectional alignment of ANNs ensures a strong security framework and facilitates the production of keys inside the IoMT network on the blockchain. (4) Mutual learning approaches synchronize ANNs, making it easier for IoMT devices to distribute synchronized keys. (5) XGBoost and ANN models were put to the test using BoT-IoT datasets to gauge how successful the suggested method is. The findings show that ANN demonstrates greater performance and dependability when dealing with heterogeneous data available in IoMT, such as ICU (Intensive Care Unit) data in the medical profession, compared to alternative approaches studied in this study. Overall, this method demonstrates increased security measures and performance, making it an appealing option for protecting IoMT systems, especially in demanding medical settings like ICUs.

**Keywords** Federated learning (FL) · Artificial neural network (ANN) · Blockchain · Machine learning (ML) · Intrusion detection system (IDS)

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## 1 Introduction

The introduction of the Internet of Things (IoT) has transformed the healthcare sector and given rise to the Internet of Medical Things (IoMT), a network of networked medical equipment and systems intended to improve patient care and healthcare delivery. But the adoption of IoMT in healthcare contexts poses significant security issues that call for creative solutions to protect private patient information and guarantee the continuous operation of medical equipment. When it comes to recognizing and reducing security risks in the IoMT ecosystem, Intrusion Detection Systems (IDS) are essential. IoMT has a huge reflect on how the medical field develops. Additionally, IoMT has grown to be a sizable and significant source of health-related information as machines

gather heart rhythm using the variety of sensors and transmit actual information to the healthcare world by using the web as a group. Shaikh and Chitre [70] proposed a method that utilizes integrated wearable technologies to continuously monitor healthcare metrics, storing the collected data in the cloud. In cases of significant issues, it promptly alerts a clinician by providing instant analysis findings. The suggested scheme lowers the expense of healthcare by the frequency of medical appointments. Rohokale et al. [64] devised a secure parenting programme using the healthcare control system for managing human healthcare parameters. The solution promotes the idea of green IoTs connectivity, saves roughly 57% of resources, and improves productivity, according to the outcomes. The centralized management of datasets causes security risks, and the storing of massive healthcare data in computer systems utilizing cloud-based servers suffers from single-point failure [71, 95, 99]. To promote the medical service, Li et al. [43] offers solutions to these issues by utilizing the advancements of blockchain techniques. Without any centralized organization in a shared ledger, it is a decentralized technology that permits transaction transfer and storing in accordance with the tasks stated in the consensus mechanism. Decentralization, anonymity, and security are just a few characteristics offered by the blockchain system. Cryptographic keys are used in blockchain storing to protect end-user credentials [36, 67]. A blockchain-based edge mechanism for information theft and information confidentiality of the sick person's health related record was proposed by Aujla and Jindal [7]. In respect of block time to prepare, headers purpose of preparing, tensor reduction ratio, and estimation mistake, the subsystem analysis demonstrates its performance. Blockchain technique acceptance IoTs has many advantages, including constancy, visibility, and information traceability [93]. The medical industry has numerous blockchain-based IoMT applications, like the handling of EMR [72], remote patient tracking [31], and medication tracking [65] and the resist against communicable diseases [91]. IoT devices are guaranteed to be vulnerable to hacking and capable of being managed centrally to create IoT botnets [81, 94]. Significant data leaks, violations, and infringements occur as a result of this botnets and assaults in the larger IoT systems [6]. Trojan, Denial of Service (DoS), Distributed Denial-of-Service (DDoS), and malware threats are few of the drastic effects on IoMT systems [38]. As a result, there is a lot of studies being done on privacy and authorization challenges related to IoT medical services [48]. The study of Machine Learning (ML)-based medical applications are expanding quickly. Some models for ML to have been employed in IDS for privacy concerns, whereas others have been used in the detection of disorders [16, 89,

90]. Because of the IoT devices' resources concerns, IDS approaches are essential [12].

This paper explores the area of IoMT security by putting out a unique strategy that combines the strength of coordinated key exchange methods, Artificial Neural Networks (ANNs), and federated cloud-based ML. Through a coordinated effort between these cutting-edge technologies and blockchain, which acts as a secure basis for data integrity and access control, the main goal is to strengthen the security architecture of IoMT.

ANN integration makes it easier to create synchronized, reliable encryption keys [24, 25, 68], and federated cloud-based ML makes intrusion detection dynamic and effective. The security, integrity, and accessibility of medical data are ensured by this unified architecture, which provides a multi-layered defense mechanism against possible attacks inside IoMT.

This paper proposed a paradigm for integrating Federated Learning (FL)-based IDS networks and blockchain technique into IoT security in medical services. The architecture can aid with appropriate model accumulation, keep safe record of transactions of the local model weights, and efficiently defend the systems against poisoning assaults.

The principal issues that require attention are, in brief, as follows:

- To confront FL poisoning attack, further practical techniques and investigations are needed.
- The shortcomings of IoMT devices expose them to hacking and control from a centralized place, which can result in IoT botnets. Serious data breaches, violations, and infringements occur as a result of these assaults and botnets in the larger IoT networks.
- Another challenges like botnets, DDoS, DoS, and trojans have a significant impact on IoMT systems. The privacy and permission issues relating to IoMT medical services must thus be addressed.
- Traditional cloud computing models' inefficiencies and excessive delay are brought on by the congested transmission channel, which is another challenge.
- Confidentiality is compromised when personal information is uploaded to a centralized server, although it is vital to do so in order to train the central model. This one failure point compromises both the operations' standards and the confidentiality of the information. This is another challenge need to be addressed.
- The primary IDS have a significant delay. The expense associated with gathering various types of data inside the 5 G/6 G facilities including texts, speech, movies, and virtual reality and augmented is rather substantial.

- Despite the FL's improvements, it still faces a formidable challenge in the form of the poisonous attack on the training data by the global model. Weights added together—that is a MITM attack. Employing this inverse model The personal information may be compromised with the combined weights of local models and engineering.
- To protect the transfer of cryptographic keys in blockchain a reliable key exchanging technique is needed.
- In order to counter both basic and genetic assaults, it is necessary to implement a cryptographic key swap-over method.
- An approach to organize the neural key exchange process has to be devised in order to enhance effectiveness and confidentiality.

The study's overarching goal is to strengthen the healthcare industry's IoMT security by combining two approaches:

- The FL-guided IDS safeguards against cyberattacks by collectively identifying abnormalities in the network without exchanging confidential patient information.
- The Neural Key Exchange Mechanism facilitates a safe method for IoMT devices to exchange encryption keys, ensuring the protection of data.

Ensuring robust security measures for IoMT is imperative due to the collection and transmission of highly confidential patient data by these devices. The reason is as follows:

**Data privacy** Breaches have the potential to reveal medical information, which can result in identity theft and discriminatory actions.

**Patient safety** Manipulated data might result in incorrect diagnosis and treatments.

**System disruption** Attacks have the potential to interrupt vital medical equipment, therefore endangering patients.

The proposed technique tackles these problems by:

**Safeguarding data privacy** FL enables training on individual data without the need to share it, hence minimizing the potential for security breaches. Edge-based intrusion detection allows for early threat identification, resulting in a quicker reaction to attacks.

**Enhanced encryption** The neural key exchange method establishes highly secure communication channels.

The major goal of this study is to create an IoMT ecosystem for healthcare that is very reliable and secure.

The benefits of our suggested system include:

- To use a light-weight detection algorithms to deal with edge nodes' resource constraints and limited storage, this paper proposed an IDS model for stopping threats on

medical systems. The ANN was chosen because to its benefits, including its enhanced quality with the diversity of facts in IoT, like ICU in medical services, and lack of limits on datasets and dissemination.

- The implementation of an edge-cloud IDS structure to avoid centralized issues like single points of failure and safeguard the confidentiality of locally learned data, that is required for a number of applications in the medical industry.
- In addition, using the detection algorithm at the network edge close to the threat origin speeds up the detecting responses and lessens the stress on the cloud.
- Combining blockchain techniques with FL in this way prevents infecting threats, enables total visibility and data integrity over the decentralized training phase, and stores nearby weights for adjusting the global model.
- For generation of key in blockchain ANN based synchronization is used.
- The crucial components are the uniform weights acquired during ANN synchronization in IoMTs. This approach is being used to create cryptographic keys over an open network.
- The time it takes for the Anti-Hebbian and Random Walk learning coordination algorithms in the suggested technique is significantly lower than that of existing approaches.
- The key interchange techniques that are outlined by Dolecki and Kozera [17], Sarkar [66], Jeong et al. [32], Dong and Huang [18], and Teodoro et al. [77] were investigated in this article. Their shortcomings were investigated also within this study.

Here is the structure of the article: The second section is devoted to the literature review. In Sect. 3, we present the proposed technique. Section 4 discusses several assaults on neuronal key synchronization. In Sect. 5, we discuss about how to implement the suggested approach. In Sect. 6, we can see the outcomes and the analyses. This concludes Sect. 7 of the article.

## 2 Related works

Conventional, centralized IDS examine network data to identify any malicious behavior. Nevertheless, this strategy gives rise to privacy problems in the IoMT as it frequently necessitates the gathering and retention of delicate medical information [58]. ML and DL are potential options for detecting anomalies in IoMT network traffic. Nevertheless, the process of training ML/DL models effectively often requires extensive datasets, which might pose a challenge

to privacy due to the sensitive nature of IoMT data [58]. FL provides a method for training IDS models in the IoMT while maintaining privacy. FL enables devices to collectively train models using their respective data without the need to share the raw data. This ensures the confidentiality of patient information while facilitating the creation of powerful intrusion detection system models [97]. Blockchain technology has several advantages for improving the security of the IoMT. Blockchain technology enables the creation of an immutable data ledger, which securely records system activities. This makes it highly challenging for attackers to manipulate data or carry out replay assaults [87]. This is especially pertinent in the context of the IoMT, since a hacked central server has the potential to expose a significant number of devices. Blockchain technology can provide a safe method for storing and distributing cryptographic keys needed for encrypting data in IoMT systems [82]. By utilizing a decentralized approach, the method suggested by Dhasaratha [14] contributes to the improvement of clinical surveillance, as well as the guarantee of safe communication and complete data privacy. In a decentralized setting, the primary goal is to enhance the effectiveness and flexibility of the reinforced FL process while simultaneously maintaining the confidentiality and safety of data for IoMT systems via the use of Blockchain technology.

A cloud-based approach that transports and saves clinical information was suggested by Mohapatra et al. [57], and its access control incorporates accepting credentials through a bar code scanner. The physician can safely check health information and provide insightful commentary. A core service named cloud is made available by cloud computing technology to handle and keep the information plan to learn the ML model. LUCID, a Convolutional Neural Network (CNN)-based IDS, was suggested by Doriguzzi-Corin et al. [19] due to its capacity for recognition to distinguish between legitimate users and DDoS threats. A unique Random Neural Network (RaNN) was suggested by Latif et al. [41] for forecasting threats like DoS, malicious processing, malicious management, data type testing, espionage, and scanning. The standard ANN, the Decision Tree (DT), and the Support Vector Machine (SVM) were contrasted with the suggested RaNN. Additionally, in comparison to the competition, the suggested method RaNN show superior threat recognition rate by an overall of 5.65%. Moreover, because of the congested transmission medium, classic cloud computing models have excessive delay and inefficiencies [13]. The shortcomings of cloud computing in facilitating distributed application are addressed by edge computing [11, 51]. Edge computing is a creative technique that moves data analysis from the main network to the network edge, where it is near the final users [4, 26, 92]. Reduced delay, greater flexibility,

data confidentiality, and increased service quality are all benefits of an edge-assisted IoT layer [50]. Most previous IDS were installed in the basic cloud layer, which was unable to sufficiently fulfill the security needs for protecting data from real-time medical diagnostics. The IoT applications and systems will suffer immeasurable harm if the infiltration is not discovered timely [54, 75, 86]. Uploading private details to a centralized body compromises confidentiality, yet it is necessary to do so to train the central model [27, 49, 74]. The confidentiality of the information and the standard of the operations are compromised because of this only failure point [33, 83]. The centralized IDS require a lot of time. In the 5 G/6 G infrastructure, gathering a various of data types—like texts, voice, and AR/VR, videos—is extremely expensive [1]. Because of earlier difficulties, studies on medical service privacy remains in its infancy, and the application of intruder sensing technologies has prompted more stringent standards. Although ML and Deep Learning (DL) technologies significantly aid in resolving practical issues, they are subject to a number of limitations [23, 45, 47]. Removing the training information scattered throughout the portable devices, having trained portable designs nearly at the edge layer, grouping naively calculated notifications within the server, acquiring knowledge advertisement learning alerts to local models and finally the global model, McMahan et al. [56] with the team Google that provided substitute method to the centralized learning of Deep Neural Network (DNN). This strategy was known as FL. For a number of technology in the medical industry, this strategy ensures the security of the locally training dataset. Zhao et al. [98] devised a multitask DNN. CICIDS 2017, ISCVPN 2016, and ISCXT datasets were used for examination. The outcomes demonstrated that the proposed approach reduces the cost of preparation time as much as centralized training and has a decent prediction accuracy in multi-tasks. To work around the limitations of IoT, the suggested model must optimize the DNN architecture.

Two FL algorithms with ANN and Logistic Regression (LR) were suggested by Rajendran et al. [62] to protect the confidentiality of user information in medical services. FL delivers higher in correctness and confidentiality relative to ANN models. The effect of FL on the development of digital technology was examined by Rieke et al. [63] medicaid by outlining the difficulties around the confidentiality of medically personal information without the necessity to centralize or share information. Although the FL's advancements, it faces a significant obstacle, that is the global model's poisonous assault that targets the training data? Combined weights, that is a type of middleman assault. Used this inverse model with combined weights from local models, engineering, the personal information

could to be undermined [34]. The researchers of [59] developed a brand-new FL incorrectly modifies the learning process that categorizes dangerous traffic, hence poisoning the gateway into the combined information for the local IDS models. When a malicious actor compromises the training sets by leaking even minute amounts of harmful content over a gateway, this is known as an information poisoning attack. The result demonstrates how successfully the assault damaged the information. A more potent model poisoning assault than the dataset poisoning attack was suggested by Bagdasaryan et al. [8]. Accumulated model upgrades that train the global model after averaging are impacted by the suggested poisoning attempt. To contaminate the model when it is being trained and the personal information is being copied, Zhang et al. [96] developed Generative Adversarial Networks (GANs). Following training, the created model information's labels is tainted, increasing the impact on the worldwide models. According to Nguyen et al. [59], the method of minimizing the product with high amplitudes of notifications to average out tainted information could harm the efficiency and have a detrimental effect on the model's primary objective. A decent fix might be offered by integrating new techniques with the FL, like blockchain, lightweight DL, and contemporary communication methods and encryption standards [1]. The advantages of utilizing blockchain technique for IoT systems in medical applications have been extensively studied. A deep blockchain architecture was suggested by Alkadi et al. [3] for IDS-based IoT networks to find intrusions in the centralized virtual infrastructure. Although the suggested architecture does more over inference and data contamination threats, confidentiality issues remain because of centralized training. Problems with dealing with varied and unstructured medical terminology might affect the accuracy of Li et al. [46]'s work on information-enhanced Text2SQL interpretation in complicated electronic health records. Wang et al. [84]'s umbrella review on case management for cancer patients faces limitations due to heterogeneity among included studies, which may introduce biases and affect the conclusiveness of the findings. For online review spammer identification, Wu et al. [88]'s semi-supervised probabilistic collaborative learning model could struggle in situations where labelled cases are hard to come by. Huang et al. [28]'s systematic review and meta-analysis comparing PET/CT and PET/MRI for detecting biochemically recurrent prostate cancer may have limitations inherent to the included studies, potentially affecting the robustness of the conclusions. It may be difficult to strike a balance between privacy and utility when using differential privacy for secure consensus management of cooperative-competitive multi-agent systems, as proposed by Ma and Hu [55]. Intermittent connectivity may provide challenges to the synchronization

of Wang et al. [85]'s wide-area heterogeneous multi-agent systems operating across clustered networks. Li et al. [44]'s investigation into nursing students' risk perception during the COVID-19 pandemic may have limitations in generalizability, and the mediating and moderating effects may be influenced by various contextual factors. Reporting biases and inadequate representation of specific incidents may have affected the results of Cai et al. [10]'s study on medical professional violence in China, which relies on national judgment papers. Possible computational cost and efficiency constraints may restrict the use of Liu et al. [53]'s federated neural architecture search to ensure the security of medical data. Overall, these limitations highlight the diverse challenges researchers encounter in their respective domains, emphasizing the ongoing need for refinement and innovation in methodologies and techniques across various fields.

The benefits and shortcomings of the current methods are summarized within Table 1.

Based on the aforementioned findings, further workable strategies and investigations are required to address FL poisoning assault.

### 3 Proposed methodology

The proposed approach in the research combines two techniques to enhance IoMT security:

#### ***Complex-valued ANN-based key exchange mechanism***

This creates a secure way for IoMT devices to exchange encryption keys for data protection.

***Federated learning-guided intrusion detection system (FL-IDS)*** This focuses on collaboratively detecting anomalies in the network traffic without sharing the raw patient data.

Let's break down how these work together:

#### **Complex-valued ANN-based key exchange**

***Device pairing*** When two IoMT devices need to communicate securely, an ANN is used to establish a shared secret key.

***Mutual learning*** Both devices participate in a training process where the ANN learns a specific pattern based on their locally generated data.

***Key derivation*** The trained ANN on each device generates a key based on the learned pattern. Ideally, due to the mutual learning process, both devices will end up with the same key.

This approach provides several advantages:

***Stronger security*** The key is derived from a complex, learned pattern, making it difficult to crack.

***Decentralized management*** No central server is needed for key generation, reducing reliance on a single point of failure.



**Table 1** Limitations and characteristics of front-line methods

| Authors               | Identified issues                                                                                                                                                      | Research findings                                                                                                                                                      | Shortcomings                                                                                                                                                                                 |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Latif et al. [41]     | It was investigated how FL affected the advancement of digital technology                                                                                              | Without the need to centralized or exchange information, the issues surrounding the confidentiality of medically personal information were highlighted                 | Additional practical strategies and research are required to counter the poisoning attack on FL                                                                                              |
| Nguyen et al. [59]    | Created a new FL poisoning gateway by incorrectly changing the learning algorithm that classified hazardous traffic into the combined information for local IDS models | The outcome illustrates how effectively the information was harmed by the assault                                                                                      | Data poisoning occurs when an adversary taints the training sets with trace levels of harmful substances introduced through a gateway                                                        |
| Alkadi et al. [3]     | The benefits of using blockchain technology for IoT applications in the medical field                                                                                  | IoT networks based on IDS can be used to detect intrusions into a centralised virtual infrastructure                                                                   | Data poisoning and inference problems persist as a result of centralized training, which raises issues regarding confidentiality                                                             |
| Rajendran et al. [62] | Safeguard the confidentiality of user information in medical services                                                                                                  | LR and ANN-based FL algorithms were proposed                                                                                                                           | Suffers from MITM attacks                                                                                                                                                                    |
| Zhao et al. [98]      | Protect the privacy of patient data when providing medical services                                                                                                    | The results showed that the recommended method has a respectable multi-task prediction accuracy and decreases preparatory time costs similarly to centralised training | The recommended model must optimize the DNN architecture in order to get past IoT constraints                                                                                                |
| Memahan et al. [56]   | Privacy is still in its infancy, the use of intruder detection technology has led to stricter regulations                                                              | Different method from the DNN's centralized learning                                                                                                                   | There are a variety of downsides to ML and DL technologies, despite the fact that they greatly aid in solving real-world issues                                                              |
| Agrawal et al. [1]    | Confidentiality is jeopardised when personal information is uploaded to a central location, yet it is essential in order to train the central model                    | This one failure point compromises both the operations' standards and the confidentiality of the information                                                           | Lots of time is required by the core IDS. The 5 G/6 G infrastructure makes it expensive to gather many types of data, including texts, audio, videos, and augmented and virtual reality data |
| Wang et al. [86]      | The primary cloud tier was where the majority of the previous IDS was installed                                                                                        | The safety standards for protecting data from real-time medical diagnostics were not sufficiently addressed by this                                                    | If the intrusion is not identified in a timely manner, the IoT applications and systems would sustain irreparable damage                                                                     |
| Lim et al. [50]       | Offers FL in edge network                                                                                                                                              | An edge-assisted IoT layer has the advantages of decreased latency, more flexibility, data confidentiality, and improved service quality                               | Lack of trust                                                                                                                                                                                |
| Alwarafy et al. [4]   | Security issues in edge computing                                                                                                                                      | A cutting-edge technique called edge computing allows data processing to be moved from the core network to the network's edge, which is located near to the users      | Suffers from privacy                                                                                                                                                                         |

**Dynamic keys** The keys can be constantly updated as the devices interact and the ANN patterns evolve, offering enhanced security against replay attacks.

#### FL-IDS

**Local training** Each IoMT device trains a local intrusion detection model on its own data. This local model analyzes and identifies potential anomalies in the device's network traffic.

**Federated learning** The model updates (weights) from each device are sent to a central server. These updates don't contain the raw data itself, but rather the learnings from analyzing the local data.

**Central aggregation** The central server aggregates these model updates using a secure aggregation function. This aggregation process aims to combine the learnings from all devices while ensuring privacy by not revealing individual device data.

**Global model update** The aggregated update is then sent back to all the devices.

**Local model improvement** Each device updates its local intrusion detection model with the received global update, improving its ability to detect anomalies.

This FL-based approach offers several benefits:

**Privacy preservation** Raw patient data remains on the device, minimizing privacy risks.

**Improved detection** Collaborative learning allows the IDS to adapt to diverse attack patterns across the network.

**Reduced communication overhead** Only model updates are transferred, reducing bandwidth usage for resource-constrained devices.

This combined approach aims to create a robust and privacy-preserving security system for IoMT networks.

The paper explores the use of complex-valued neural synchronization for key generation in blockchain, namely in Sects. 3.1 and 3.2 discusses the intrusion detection strategy based on FL in blockchain.

### 3.1 Complex-valued neural synchronization-based key generation in blockchain

One particular example of an online learning situation is the synchronization of ANN. Two neural networks begin with randomly-generated weight vectors. At each stage, they utilize a shared input vector, compute their outcomes, and exchange information with each other. When they reach an agreement on the relationship between the current input and output, their synaptic weights are adjusted based on the applicable learning protocol. In the event of discrete weight values, this technique results in complete synchronization in a limited number of steps.

Furthermore, even after acquiring further knowledge to implement rules, the weights that are being compared in both networks have the same magnitude. Complete synchronization is therefore an absorption condition. In addition, a third neural network may be trained utilizing the examples, input vectors, and output values produced by the synchronization method. This neural network is similar to a student network that, due to its inability to affect others, strives to acquire knowledge about a time-dependent input–output relationship. The average number of stages required for synchronization and for learning is the same for perceptrons, which are the most basic neural networks. There is therefore little difference between these two circumstances. The more complex ANNs, however, exhibit an unexpected phenomenon: two neural networks that learn from one another synchronize more quickly than a third network that merely involves hearing the communication.

To address the key-interchange issue in blockchain, leveraging the contrast between unidirectional and bidirectional contact proves beneficial. Thus, partners A and B synchronize their Artificial Neural Networks (ANNs). This enables them to expedite the construction of their shared session key compared to an adversary, who might be training a distinct neural network. Consequently, the efficacy of the neural key-interchange protocol's privacy relies on the disparity between synchronization and learning.

#### Mutual learning for key agreement

**Device pairing** When two IoMT devices need to communicate securely, they initiate a key exchange process.

**Local data generation** Each device generates its own random data specific to the pairing session.

**ANN training** Both devices use this local data to train their respective ANNs. The goal is for both ANNs to learn a similar pattern based on the unique data from each device.

**Key derivation** Once trained, each device's ANN utilizes the learned pattern to generate a key. Ideally, due to the similar training data and the mutual learning process, both devices will end up with the same key.

#### Benefits of this approach for synchronized key distribution

**Dynamic keys** The keys are derived from a complex, learned pattern that constantly evolves as the devices interact and generate new training data. This reduces the risk of attackers using a single key to compromise multiple communications.

**Decentralized management** There is no central server involved in key generation or distribution. This mitigates the risk of a single point of breakdown and enhances the system's resilience against assaults.

**Implicit synchronization** Through the mutual learning process using the same random data, both devices inherently arrive at the same key, achieving synchronized key distribution.

In this situation, ANNs are multiple-layered feed-forward networks employed in neural cryptography by both partners and attackers. Three layers are in ANN: input, output and concealed. Input layer comprises  $L \times M$  input nodes. Hidden layer comprises  $L$  concealed nodes.

Each neuron bears resemblance to a distinct perceptron. The output layer is comprised of a single output neuron. The weight is a complex number that may be represented by Eq. 1.

$$v_{q,j} = d_{q,j} + e_{q,j}i \quad (1)$$

$d_{q,j} \in \{-l, -l+1, \dots, l\}$ ,  $e_{q,j} \in \{-l, -l+1, \dots, l\}$ ,  $q = 1, 2, \dots, W$ ,  $j = 1, 2, \dots, M$  denotes the network's  $q$ th concealed layer,  $l$  is the synaptic depth, and represents the vector components.  $y_{q,j}$  serve as the input vector, it has the complex definition that follows:

$$y_{q,j} = d_z + e_z i \quad (2)$$

where  $d_z \in \{-1, 1\}$ ,  $e_z \in \{-1, 1\}$ . The  $q$ th hidden layer can be depicted by Eq. 3.

$$g_q = \frac{1}{\sqrt{M}} \left( \sum_{j=1}^M (d_{q,j} d_z) + \sum_{j=1}^M (e_{q,j} e_z) \right) \quad (3)$$

and the weight can be utilized to swap a pair of group keys in single coordination method as it comprises both real and imaginary parts (Eqs. 4 and 5).

$$g_q = \frac{1}{\sqrt{M}} \left( \sum_{j=1}^M (d_{q,j} d_z - e_{q,j} e_z) + i \sum_{j=1}^M (d_{q,j} e_z + e_{q,j} d_z) \right) \quad (4)$$

$$\alpha_q = d_{\alpha_q} + e_{\alpha_q} i \quad (5)$$

where  $d_{\alpha_q} = \text{sgn}(\text{Re}(g_q))$  and  $d_{\alpha_q} = \text{sgn}(\text{Im}(g_q))$  the result function can look like this (Eq. 6).

$$\chi = \prod_{l=1}^q (\text{Re}(\alpha_q)) + \prod_{l=1}^q (\text{Im}(\alpha_q)) \quad (6)$$

In comparison to Eqs. 3 and 4, this concealed layer does not increase safety but lengthens the synchronization operation as  $e_q e_z$  and  $d_q d_z$  may trigger a rise in step during synchronization operation in an undesirable manner.

Let  $D$  and  $E$  are trying to secure a private key through an open network. Complex valued numbers can be found

in  $D$  and  $E$  both. After that, the mutual training mechanism appears in the following form:

The weights  $v_q^D$  and  $v_q^E$  are each first set to a complex random quantity by  $D$  and  $E$ .  $D$  and  $E$  accept a shared input vector  $y_{q,j}$  which is produced at every stage randomly and calculate the result  $\chi^{D/E}$  by Eqs. 3–6.

After that, on the open link,  $D$  and  $E$  interchange  $\chi_D$  and  $\chi_E$  with one another. When  $D(E)$  gets  $E(D)$ ,  $D(E)$  alters the synaptic weight in accordance with the guidelines beneath.

- if  $\chi_{re}^D \neq \chi_{re}^E$  and  $\chi_{im}^D \neq \chi_{im}^E$  then the weight  $v_q^{D/E}$  cannot be updated.
- if  $\chi_{re}^D = \chi_{re}^E = \text{Re}(\alpha_q^{D/E})$  and  $\chi_{im}^D = \chi_{im}^E = \text{Im}(\alpha_q^{D/E})$  then modify the weight of  $v_{q,j}$  of  $\alpha_q^{D/E}$
- if  $\chi_{re}^D = \chi_{re}^E = \text{Re}(\alpha_q^{D/E})$  and  $\chi_{im}^D = \chi_{im}^E \neq \text{Im}(\alpha_q^{D/E})$ , then modify the real portion of the synaptic weight  $v_{q,j}$  of  $\alpha_q^{D/E}$
- if  $\chi_{re}^D = \chi_{re}^E \neq \text{Re}(\alpha_q^{D/E})$  and  $\chi_{im}^D = \chi_{im}^E = \text{Im}(\alpha_q^{D/E})$  then modify just the imaginary portion of the synaptic weight  $v_{q,j}$  of  $\alpha_q^{D/E}$

The training principles are:

Hebbian training rule (Eq. 7).

$$\begin{aligned} v_{q,j}^{R+} &= h(v_{q,j}^R + (v_{q,j}^R \chi_{re}) \ominus (\alpha_q^R \chi_{re}) \ominus (\chi_{re}^D \chi_{re}^E)) \\ v_{q,j}^{I+} &= h(v_{q,j}^I + (v_{q,j}^I \chi_{im}) \ominus (\alpha_q^I \chi_{im}) \ominus (\chi_{im}^D \chi_{im}^E)) \end{aligned} \quad (7)$$

Anti-Hebbian learning rule (Eq. 8).

$$\begin{aligned} v_{q,j}^{R+} &= h(v_{q,j}^R - (v_{q,j}^R \chi_{re}) \ominus (\alpha_q^R \chi_{re}) \ominus (\chi_{re}^D \chi_{re}^E)) \\ v_{q,j}^{I+} &= h(v_{q,j}^I - (v_{q,j}^I \chi_{im}) \ominus (\alpha_q^I \chi_{im}) \ominus (\chi_{im}^D \chi_{im}^E)) \end{aligned} \quad (8)$$

Random-walk learning rule (Eq. 9)

$$\begin{aligned} v_{q,j}^{R+} &= h(v_{q,j}^R + (v_{q,j}^R) \ominus (\alpha_q^R \chi_{re}) \ominus (\chi_{re}^D \chi_{re}^E)) \\ v_{q,j}^{I+} &= h(v_{q,j}^I + (v_{q,j}^I) \ominus (\alpha_q^I \chi_{im}) \ominus (\chi_{im}^D \chi_{im}^E)) \end{aligned} \quad (9)$$

where  $\alpha_q^R = \text{Re}(\alpha_q)$ ,  $\alpha_q^I = \text{Im}(\alpha_q)$ ,  $v_{q,j}^{I+} = \text{Im}(v_{q,j}^+)$ ,  $v_{q,j}^{R+} = \text{Re}(v_{q,j}^+)$  and function  $h(v)$  will be utilized to ensure that every component of the  $v^{D/E}$  is in span  $[-l, +l]$  that has the following definition (Eq. 10).

$$h(v) = \begin{cases} \text{sgn}(v)l, & \text{for } |v| > l \\ v, & \text{otherwise} \end{cases} \quad (10)$$

The existing synchronization phase is then finished. This method can be performed repeatedly until the synaptic weights of  $D$  and  $E$ 's ANN are equal,  $v_q^D = v_q^E$ . The learning order is not applicable later to disrupt this synchronization



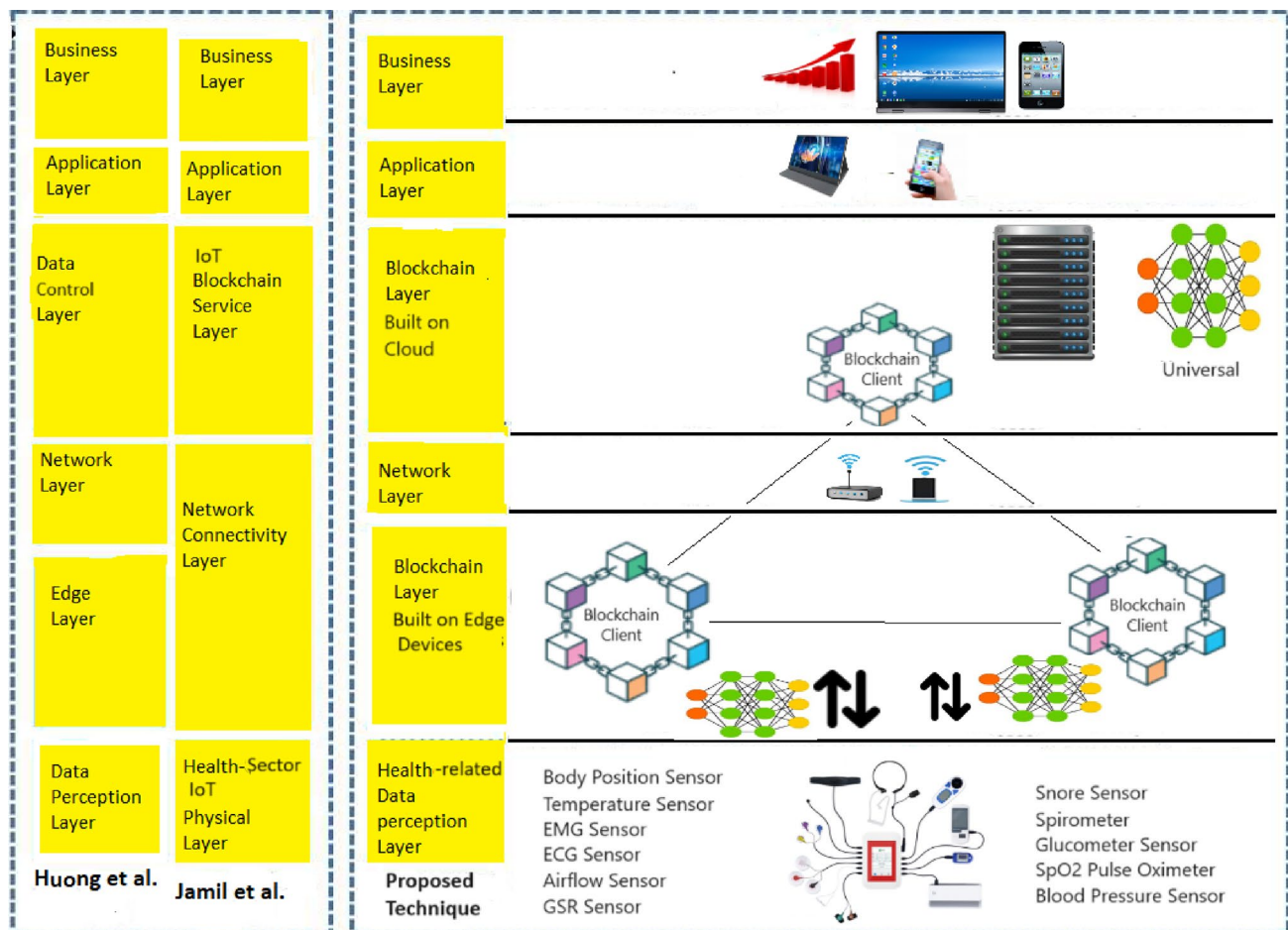


Fig. 1 The suggested framework's multilayer structure

because the movements of the synaptic weights in  $D$  and  $E$ 's neural networks are solely reliant based on the identical inputs and weights.

### 3.2 Federated learning-guided intrusion detection system in blockchain

The multi layer structures of proposed approach and those suggested in [29, 31] are depicted in Fig. 1. A physical layer, a connection layer for networks, a cloud layer for the IoMT' blockchain technology, an application layer, and a business layer make up the suggested system design in [31]. The suggested solution uses blockchain-based smart contracts to track important parameters of patients excluding classifying the data used by an ML algorithm. The detection system for intrusions made no mention of security concerns. The system is centrally located,

possesses a single failure point, and compromises patient data as it transmits it to a main server.

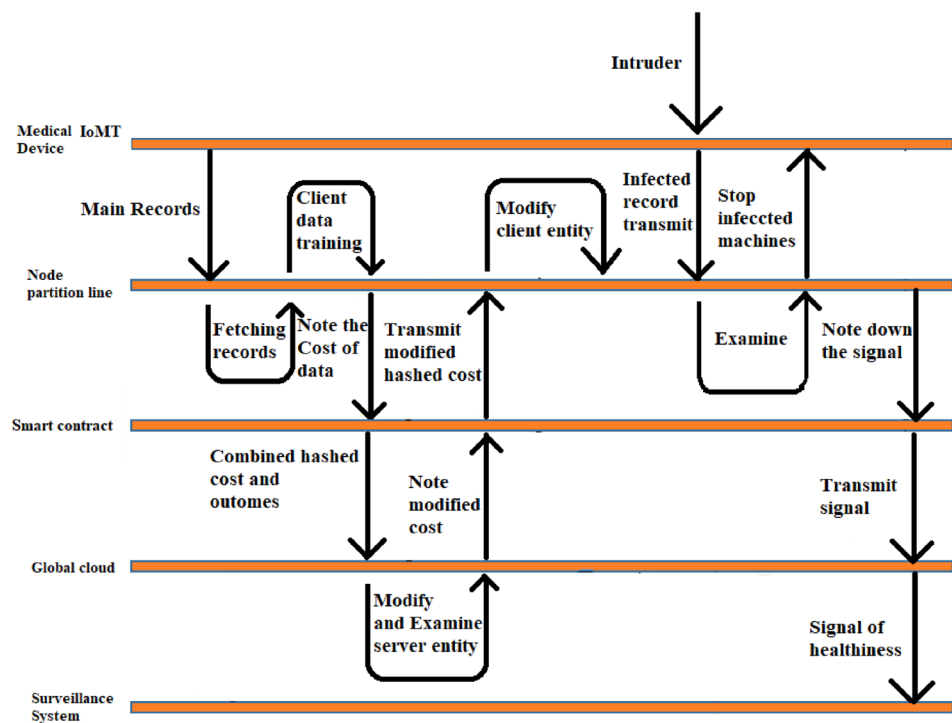
Huong et al. [29] proposes a framework architecture consisting of several layers: data sensing, applications, edges, cloud, network, and business.

The suggested scheme is a federated multi-attack edge level detector. When transferring the weights for accumulation to the server, the IDP (Intrusion Detection Program) did not address the issue of poisoning assaults. The precise results have a 92% mean, which could be improved.

The following is a description of the proposed framework's structural layers:

1. The patient information perception level encompasses IoT sensors equipped for the ICU. Devices in intensive care unit cases can be categorized into two groups: those

**Fig. 2** The proposed system's workflow execution



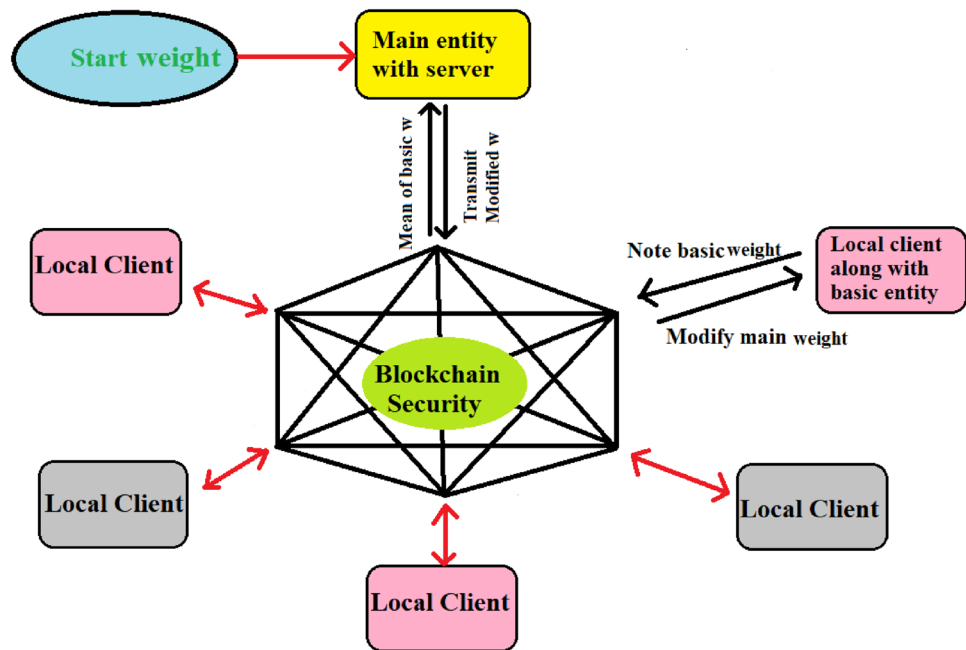
monitoring the patient's condition and those monitoring the ward's surroundings.

2. The blockchain level, established on the perimeter, comprises IoT gateways. Several medical detection systems are present at every gateway. Since there is no universal inner interface for physical health sensing, the gateways allowed a wide range of network access protocols. A multi-attack screening is carried out by an IoT gateway. An easy-to-use IDS was created at the Expert System (ES) to standardize their information and stop various ANN-based assaults. If a specific assault occurs, the suggested component will be taught at the network edge and evolved in FL mode to defend the cloud or other assets by blocking the flow from its gateways. Even as assets for an assault draw closer, the time required of an infiltration will decrease. In furthermore, since the FL model works with smaller samples, there will be fewer computer and computing power available. The heaviness of every internal model are uploaded to a blockchain-distributed record when the module learning phase is finished and are then kept within chained blocks that link gateways sites with the following cloud tier's servers. These interconnected blocks will also be used for average computation and combinational purposes. Last but not least, because it is driven by consensus mechanisms (smart contracts), the network is protected by a cryptic hash function that unites the net-

work's nodes and is hence impervious to alteration. Figure 2 details how the entire system's workflow is executed. One of FL's main significant issues, poisoning threats, is addressed by the proposed approach. Within which each ES obtains the latest weighting factors, encodes the information gathered, and creates the associated signature to use a secret key unique to it. The active blockchain level is managed by a smart contract, and ES then combines the encrypted message and sends it along with the signatures while maintaining control over information confidentiality and integrity. By using shared key of the ESs and the information recorded as blockchain hashed blocks in accordance with the guidelines of the shared ledger, the shared ledger validates the legitimacy of such messages whenever getting the information from all ESs. The CPCC (Central Public Cloud Center) can get the consolidated plaintext by utilizing one's own secret key to access the saved blocks from the enabled blockchain. The interactions between ESs and CPCC, as well as ES and the appropriate local model, are both 2 different in most IoT edge application domains. Similar to data storage centered on endpoint edge computing, native model can transfer and download information to or from CPCC using blockchain networks and ES.

3. Ensuring the security of data transmissions from lower to higher levels is the responsibility of the network layer.

**Fig. 3** Proposed technique's weight aggregation into the blockchain network architecture



Connectivity at this level is considered essential for route management.

4. Blockchain level is powered by the cloud. The cloud manages the accumulated weight-values from the ESs in the blockchain, calculating mean weight-values, and modifying the global weight-values of the ANN algorithm. To effectively safeguard the network, the cloud regularly distributes the aforesaid modified weight-values to all gateways for modifying local model. The proposed structure of the blockchain network's accumulation of regional and global weight-values is shown in Fig. 3.
5. Important notes for medical services are monitored at the application layer.
6. Based upon information that has been evaluated and obtained from bottom layer, business layer assists the organization of the entire medical application system in creating business strategies, flow diagrams, and executive summaries.

#### Drawbacks of the existing method in the absence of blockchain

- **Central aggregation point** The FL-IDS depends on a central server to collect and combine model changes. This server is susceptible to possible attacks.

#### Potential advantages of integrating blockchain:

- **Decentralized aggregation** Blockchain has the potential to replace the central server used for gathering model changes in the FL-IDS. This would remove a solitary instance of a malfunction and enhance the system's resistance to tampering.
- **Secure key storage and management** Blockchain can provide a safe method for storing and distributing keys for the ANN-based key exchange mechanism. Utilizing this approach can significantly augment the security of key management in comparison to merely depending on the devices themselves.
- **Immutable audit trail** The blockchain records all transactions, including model modifications and key swaps, in an immutable manner. This can enhance the transparency and auditability of the security system. Difficulties in incorporating blockchain technology:
- **Scalability** Blockchain transactions may incur high computing costs and exhibit poor processing speeds. This may not be optimal for real-time intrusion detection in IoMT systems.
- **Complexity** The integration of blockchain introduces a new level of intricacy to the system, necessitating further development and resource management.

Following Algorithm 1 describe the FL-guided intrusion detection procedure.

**Algorithm 1:****Input:**

$P$  denotes the numbering of nodes in FLIDS;  
 The total number of global iteration is denoted by  $h$ ;  
 Number of epoch is denoted by  $D$ ;  
 The size of batch in local is symbolized by  $N$ ;  
 Total number of gateways for the edge is symbolized by  $L$ ;  
 $p_l$  denotes the amount of information at edge gateway  $l$ ;  
 and the rate of learning is denoted by  $\delta$ .

**Output:** Modified Weight  $W_t$ **Procedure:**

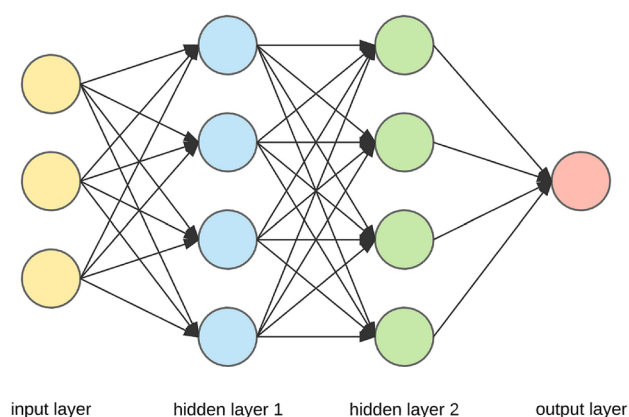
Updataion of Node1 at Server :  
 Initialization of weight value  $w_{t0}$   
 //The Blockchain network's is denoted by FLIDSCreation of FLIDS at Node 1  
 Using the Internet Protocol address establish the connection with FLIDS at Node1  
 For each edge(local) from 1 to  $D$  do  
 //FLIDS Node number = Node number at edge + 1  
 $P = D + 1$   
 Authorize mining of other nodes (transmitting and modifying) at Node 1  
 Using the Internet Protocol address establish the connection with FLIDS at Node  $N$   
 End For  
 Publication of  $w_{t0}$  (initial weight)  
 For each epoch (global)  $h = 1, 2, \dots$  do  
 For each node  $P \in \text{FLIDS}$   $P$  from 2 to  $(l + 1)$  in parallely do  
 $W_{t_{h+1}} \leftarrow \text{UpdationOfNodesAtEdge}(\text{Node } P, W_{t_h})$   
 End For  
 $W_{t_{h+1}} \leftarrow \sum_{L=1}^L \frac{p_l}{p} W_{t_{h+1}}$   
 $p = \sum_l p_l$   
 $W_t \leftarrow W_{t_{h+1}}$   
 Publication of  $w_{t0}$  (modified global weight) at FLIDS  
 End For  
 End procedure  
 Procedure : UpdationOfNodesAtEdge (Node  $P$ ,  $W_{t_h}$ )  
 $N \leftarrow$  (splited data  $p_t$  to a batch size of  $N$ )  
 //training localized models for feed – forward propagation at the edge  
 For each epoch (local)  $D = 1, 2, \dots$  do  
 //Use SGD (Stochastic Gradient Descent) in Backpropagation to update the weights (local)  
 $W_t \leftarrow W_t - \delta \nabla_{gl}(W_t)$   
 // $\nabla_{gl}(W_t)$  is the mean gradient of data (local)  
 End For  
 Publication of  $w_t$  (modified local weight) in FLIDS at Node  $P$   
 End procedure

The human brain served as the inspiration for ANN, a concurrent, decentralized system [73]. The Multilayer Perceptron (MLP) is the most popular ANN paradigm [15]. An ANN was employed for the suggested IDS. Based on the information obtained from the BoT-IoT, the feed-forward neural network is trained by the utilization of a back propagation approach, as seen in Fig. 4. The hyper-parameters utilized in the suggested prediction model are listed in Table 2.

The proposed scheme makes the following predictions in test cases:

1. The similar architecture and hyper-parameters are shared by all internal models, however they are learned using various original dataset partitions.
2. The same starting weights are used for all clients.
3. Clients receive the weight changes simultaneously and without respect to whether they took part in the most recent global period.
4. All clients learn at the same rate.

**Patient level (data generation)** Diverse IoMT devices (wearables, implantables, and sensors) generate continuous



**Fig. 4** Architecture of an ANN (binary classification)

**Table 2** Hyper-parameters incorporated into the suggested detection model

| Hyper-parameters       | Value                             |
|------------------------|-----------------------------------|
| Total number of epochs | 2 : 10(+1)                        |
| Group size             | 100 : 1000(+100)                  |
| Learning rate          | 0.001 : 0.1(+0.01)                |
| Type of classification | Binary                            |
| Optimization algorithm | Stochastic gradient descent (SGD) |
| Activation function    | Sigmoid                           |

health data. Pre-processing or edge computing anonymizes data where possible, removing direct identifiers. A clear, granular informed consent mechanism is built into devices or patient-facing apps.

**Local healthcare institution** FL nodes act as secure data enclaves. No raw data was exposed externally. The local intrusion detection model trains on decentralized data within the institution. There is potential for differential privacy at this level to provide additional protection.

**Federated aggregation server (trusted third party)** Receives encrypted updates from FL nodes; there is no direct access to data. Aggregates updates to improve the global intrusion detection model without revealing individual patient information. Potentially explores homomorphic encryption for computation on encrypted updates.

**Key exchange layer** The neural key exchange protocol leverages shared model parameters, enhancing secure communications between IoMT devices and FL nodes. Continuous key rotation to reduce vulnerability.

### Shared ledger through model weights

**Local training** Each IoMT device trains its own local intrusion detection model on its data. This training process

generates “weights” that reflect the importance of different features in identifying anomalies.

**Weight sharing** These local model weights, instead of raw patient data, are uploaded to a central server.

**Secure aggregation** A secure aggregation function combines the weights from all devices. This function ensures privacy by not revealing individual device weights but capturing the collective learning across the network.

**Global model update** The aggregated weights, representing the combined knowledge from all devices, are sent back to each device.

**Local model improvement** Each device updates its local model with the received, aggregated weights, improving its ability to detect anomalies.

This process creates a shared ledger in the form of the continuously updated, aggregated weights. These weights represent the collective knowledge of the network about potential attack patterns without revealing any individual patient data.

### Benefits for data visibility and integrity

**Improved anomaly detection** By aggregating knowledge from all devices, the FL-based IDS can adapt to diverse attack patterns across the network, potentially leading to better anomaly detection.

**Reduced risk of poisoning attacks** Malicious actors might try to manipulate the training data to weaken the IDS. FL mitigates this risk by only sharing weights, making it harder to inject misleading information.

**Enhanced data integrity** The secure aggregation function helps ensure that the combined weights accurately reflect the collective learning and are not tampered with during communication.

### 3.2.1 Data availability and quality of training data

#### Strategies for data acquisition

**Collaboration with healthcare institutions** This work seeks to form collaborations with healthcare facilities that are willing to provide anonymized and de-identified patient data to a secure central repository for the purpose of model training and validation.

**Synthetic data generation** This work investigates the use of privacy-preserving synthetic data creation methods to produce authentic and varied datasets in compliance with ethical standards and data protection laws. Data

**Aggregation platforms:** This study explores the capabilities of FL platforms created to enable secure data sharing for model training while maintaining data privacy at the original source.

#### Ensuring data quality

**Data cleaning and pre-processing** This work employs robust data cleaning strategies to handle missing values,



inconsistencies, and outliers. It may use FL methods for distributed data cleaning.

**Data labeling processes** This research established precise data labeling standards to guarantee precise and uniform labeling of training data, which may require partnerships with medical professionals.

**Data augmentation techniques** This work allows for consideration of utilizing data augmentation strategies, including creating variations of current data points, to enhance the diversity and generalizability of the training data.

#### **Addressing privacy concerns**

**FL approach** The main advantage of the proposed solution is the FL strategy, which reduces data exchange by training models on local data at participating institutions. This effectively deals with privacy concerns by storing sensitive patient data on-site.

**Differential privacy** This work investigates the incorporation of differential privacy methods into the FL process. This process involves adding precisely adjusted random data to model updates, making it difficult to recreate the original data and enhancing patient privacy protection.

#### **Evaluation and monitoring**

**Performance metrics** The research uses appropriate performance indicators to assess the efficacy of the trained models, including accuracy, precision, recall, and generalizability to new data.

**Regular data reviews** This work allows for conducting routine evaluations of the training and validation datasets to maintain data quality and detect any biases or data drifts.

### **3.2.2 Proposed technique's potential impact across diverse fields**

#### **Potential use cases in other sectors**

**Financial services** FL can be utilized to improve security in the financial industry by preventing fraud and unwanted access to financial information, while also ensuring data privacy in decentralized networks.

**Smart grids** The technology can be utilized for secure communication and data exchange within smart grid infrastructure to safeguard against cyberattacks and ensure the dependable and effective operation of the power grid.

#### **Key elements enabling generalizability**

**FL core** The main advantage of the proposed technique is its inherent adaptability to various settings. It enables the training of models using distributed data without the need for centralized storage or sharing of sensitive information, making it appropriate for different fields with privacy issues.

**Cross-domain collaboration** This work encourages collaboration with scholars and industry specialists from other fields to assess the feasibility of the proposed system in different environments.

**Open-source development** This work allows the system to promote additional development and customization by the wider research community to tackle difficulties in many areas.

### **3.2.3 Scalability, system integration, interoperability, and sustainable system maintenance challenges**

#### **Scalability considerations**

**FL optimizations** This research investigates methods to optimize FL for large-scale implementations. These tactics consist of:

- **Hierarchical FL** This paper implemented hierarchical structures that partition data and processing over numerous layers, which decreased communication overhead and enhanced performance in FL.
- **Federated model averaging with sparsity** This paper implements Federated Model Averaging with Sparsity, which involves using methods that send only a portion of model updates during communication, leading to a notable decrease in network bandwidth usage.

**Scalable blockchain integration** This paper investigates scalable blockchain integration approaches. This involves:

**Sharding** It involves dividing the blockchain into smaller portions to enable parallel processing, which enhances scalability in extensive networks.

**Permissioned blockchains** Utilizing permissioned blockchains with a limited number of pre-approved participants saves computational overhead compared to public blockchains.

**Efficient ANN implementations** This research investigates methods to decrease the resource demands of the ANN models utilized for key exchange.

- **Model pruning** Model pruning involves eliminating unnecessary connections in an ANN, which can greatly decrease its size and computing requirements.
- **Knowledge distillation** A larger, pre-trained model can have its information transferred to a smaller, more efficient model by knowledge distillation. This process keeps the accuracy but reduces the resource utilization.

#### **Mitigating complexity**

**Modular design** The system's modular design enables separate development, deployment, and maintenance, making it easier for healthcare institutions to customize and lowering implementation complexity. Comprehensive documentation and APIs are available for each module, making integration easier and reducing development time.

**Standardization and interoperability** The paper promotes the use of standardized interfaces and protocols in the healthcare field to ensure interoperability and facilitate integration.

**User-centric design** The system's interface and administration tools focus on user-centric design concepts to empower healthcare staff with easy interfaces, comprehensive documentation, and training materials.

#### **Addressing resource constraints**

**FL with edge computing** This paper looks at how to use edge computing resources to take some of the computing load off of IoMT devices. This could make the system more scalable and reduce the need for communication in situations where resources are limited.

**Cloud-based deployment options** Investigating the practicality of utilizing cloud-based deployment alternatives to reduce strain on local resources and IT knowledge in healthcare facilities.

**Partnerships and training** Collaborative partnerships and training programs are provided with healthcare organizations and technology suppliers to alleviate resource limits and ensure successful installation and maintenance.

**Collaboration with healthcare professionals** Ongoing engagement with healthcare professionals attempts to continuously adjust the system's complexity to correspond with their workflow and resource restrictions.

#### **Model optimization**

**Lightweight model architectures** This work proposes lightweight model architectures tailored for resource-constrained devices to reduce computational complexity while preserving acceptable accuracy.

**Quantization techniques** This paper quantizes methods to decrease the accuracy of model weights and activations, which can greatly diminish memory use and computational demands.

#### **Efficient training and inference**

**Federated distillation** This technique is perfect for devices with limited resources since it reduces knowledge from a larger model to a smaller, more efficient one.

**On-device training and inference** The work investigates the possibility of conducting basic training and inference activities on IoMT devices to decrease dependence on centralized resources and network traffic.

#### **Addressing network constraints**

**Federated averaging with gradient compression** Using gradient compression in Federated Averaging helps reduce network bandwidth consumption throughout the FL process by providing compressed gradients instead of whole model updates.

**Asynchronous communication** This work implements asynchronous communication protocols that let devices engage in the FL process independently, decreasing peak network traffic and enhancing efficiency.

**Efficient communication protocols** The research explores the implementation of communication protocols tailored for resource-limited IoMT contexts. Gossip protocols and similar protocols are designed to enhance data exchange efficiency among devices and minimize network congestion.

#### **Prioritizing security and performance**

**Adapting security measures** This study focuses on security solutions that aim to strike a balance between efficacy and resource demands. It may involve modifying encryption algorithms or key lengths to reduce the computing workload while ensuring sufficient data protection. This work also includes multi-factor authentication to enhance security without compromising performance.

**Performance monitoring and optimization** This work involves implementing regulations for continual performance monitoring and optimization to ensure effective resource consumption while maintaining system security and correctness.

#### **Sustainable system maintenance**

**Federated update mechanisms** The proposed FL architecture enables efficient model upgrades through federated update mechanisms. Local institutions can download and implement enhanced models independently, reducing the need for centralized server involvement and decreasing communication overhead and potential bottlenecks.

**Open-source components** This work incorporating open-source libraries and frameworks encourages community involvement and continuous development, promoting lasting support and scalability.

**Support services** This work allows the creation of a support system to provide direction and aid to healthcare institutions that are using the system and to ensure ongoing effective implementation and troubleshooting support.

#### **Adaptability to evolving threats and technologies**

**Continuous monitoring and threat intelligence** The authors of this paper suggest updating the system on new attack vectors and IoMT vulnerabilities by connecting it to real-time threat intelligence feeds. By integrating new characteristics or retraining with relevant data, the intrusion detection model may be proactively adapted.

**Federated model updates with differential privacy** The FL configuration enables ongoing changes to the intrusion detection model while maintaining patient data privacy through differential privacy. Implementing differential privacy strategies can improve the updates, making the model more effective in defending against changing threats while also maintaining patient confidentiality.

**Environmental considerations** This work recognizes the importance of minimizing the environmental footprint of the proposed solution. Here are some potential strategies we are actively exploring:

**Energy-efficient hardware and software** Collaborating with hardware and software suppliers to utilize resource-efficient solutions for IoMT devices and server infrastructure to achieve energy efficiency.

**Optimized communication protocols** Studying communication techniques that reduce data transmission overhead and energy usage in the FL system.

**Life-cycle assessment** Performing a life-cycle evaluation to analyze the environmental effects of various system components and pinpoint opportunities for enhancement.

## 4 Security analysis

### 4.1 Attacks on neural key synchronization

A neural synchronization application is the neuronal key swap.  $D$  and  $E$  both employ an ANN with the similar architecture. The parameters  $L$ ,  $l$ , and  $M$  are all open to the public. Each ANN initiates with weights randomly selected. These original circumstances are kept hidden. The synchronization technique, which is defined in Sect. 3.1, only communicates the input matrix  $y_q$  and the total number of outputs  $\chi^D$ ,  $\chi^E$  through the public channel. As a result, each participant only knows his own ANN's internal representation  $(\alpha_1, \alpha_2, \dots, \alpha_L)$ . Protecting the privacy of this data is critical to the integrity of the key-exchange protocol. Once  $D$  and  $E$  attain complete synchronization, they utilize the weight vectors as a shared secret key.

The attacker  $A$ 's fundamental issue is that she is unaware of the internal representations  $(\alpha_1, \alpha_2, \dots, \alpha_L)$  of  $D$  and  $E$ 's ANNs. As the movement of weights is governed by the attack, its success hinges on accurately predicting the behavior of the disguised units. Major well-known attacks typically employ this strategy. However, alternative approaches exist, and it's possible that a sophisticated attack methodology could be developed, undermining the security of neural key exchange entirely. Every cryptographic mechanism is vulnerable to this kind of attack except the one-time pad. Simple assaults are covered in Sect. 4.1.1, whereas genetic attacks are addressed in Sect. 4.1.2.

#### 4.1.1 Simple attack

As per fundamental strategy, Attacker  $A$  just gives training to a 3rd ANN with input vectors  $y_q$  and output bits  $\chi^D$  as instances. These perhaps be recovered with ease by seizing the network transmissions transmitted on open network by the clients. The neural network built by  $A$  begins with random beginning weights, much as the neural networks built by  $D$  and  $E$ . The intruder determines the neural network's result at every time step. Then, the unchanged learning process is followed by  $A$  and the clients, only  $\chi^A$

is replaced by  $\chi^D$ . Therefore, updating the weights may be done using any of the subsequent equations:

HLR (Hebbian Learning Rule) (Eq. 11).

$$\begin{aligned} v_{qj}^{R_{A+}} &= h(v_{qj}^R + (v_{qj}^R \chi_{re}^D) \ominus (\alpha_q^{R_A} \chi_{re}^D) \ominus (\chi_{re}^D \chi_{re}^E)) \\ v_{qj}^{I_{A+}} &= h(v_{qj}^I + (v_{qj}^I \chi_{im}^D) \ominus (\alpha_q^{I_A} \chi_{im}^D) \ominus (\chi_{im}^D \chi_{im}^E)) \end{aligned} \quad (11)$$

Anti-Hebbian learning rule (Eq. 12).

$$\begin{aligned} v_{qj}^{R_{A+}} &= h(v_{qj}^R - (v_{qj}^R \chi_{re}^D) \ominus (\alpha_q^{R_A} \chi_{re}^D) \ominus (\chi_{re}^D \chi_{re}^E)) \\ v_{qj}^{I_{A+}} &= h(v_{qj}^I - (v_{qj}^I \chi_{im}^D) \ominus (\alpha_q^{I_A} \chi_{im}^D) \ominus (\chi_{im}^D \chi_{im}^E)) \end{aligned} \quad (12)$$

Random walk learning rule (Eq. 13).

$$\begin{aligned} v_{qj}^{R_{A+}} &= h(v_{qj}^R + (v_{qj}^R) \ominus (\alpha_q^{R_A} \chi_{re}^D) \ominus (\chi_{re}^D \chi_{re}^E)) \\ v_{qj}^{I_{A+}} &= h(v_{qj}^I + (v_{qj}^I) \ominus (\alpha_q^{I_A} \chi_{im}^D) \ominus (\chi_{im}^D \chi_{im}^E)) \end{aligned} \quad (13)$$

Therefore, despite the fact that the final output is dissimilar,  $A$  uses the inner depiction  $(\alpha_1^A, \alpha_2^A, \alpha_3^A, \dots, \alpha_L^A)$  of her personal network to predict  $D$ 's. It is clearly not the best available attack method accessible as  $\chi^D \neq \chi^A$  means that there is a minimum of one hidden unit with  $\alpha_q^D \neq \alpha_q^A$ .

#### 4.1.2 Genetic attack

The genetic assault provides an alternate strategy for the opponent, which is based on an evolutionary algorithm rather than improving the internal representation prediction. At starts with only one ANN that has been randomly started, but she can employ as many as  $N$  neural networks.

Every time the partners update the weights, the following evolutionary procedure is executed because  $\chi^D = \chi^E$  at a time step:

- $A$  finds all  $2^{L-1}$  internal representations  $(\alpha_1^A, \dots, \alpha_L^A)$  that recreate the output  $A$  as long as she possesses at most  $N/2^{L-1}$  ANNs. Following that, the learning rule is employed to update the weights in the attacking networks. In this mutation stage,  $A$  makes  $2^{L-1}$  variations of each ANN.
- It should only be necessary to keep the fittest ANN if  $A$  already has more than  $N/2^{L-1}$  neural networks. This is accomplished by effectively rejecting any networks that predicted fewer than  $F$  outputs  $\chi^D$  in the last  $C$  learning steps, with  $\chi^D = \chi^E$ . The default parameters for the selection step are a limit of  $F = 10$  and a history of  $C = 20$ .  $A$  also, maintains at least 20 of her ANNs on hand.

The algorithm that picks the fittest neural networks determines the efficiency of the genetic assault. The ANN,

which has the same internal representation sequence as  $D$ , is never discarded in the perfect scenario. Thus, the difficulty of opponent  $A$  would've been decreased to  $L$  back-propagation synchronizing, and also the genetic invasion would likely be effective. This method is not perfect, just like the other options available to opponent  $A$ .

#### 4.1.3 Adversarial attacks

It involves manipulating input data to deceive the ANN model. This research employs adversarial training methods and input validation strategies to improve resilience against such attacks. This research employs adversarial training methods and input validation strategies to improve resilience against such attacks, ultimately advancing the field of Artificial Intelligence (AI) cybersecurity.

## 4.2 Potential privacy risks in FL

### 4.2.1 Security attacks on FL and blockchain

**Data poisoning:** It involves malicious individuals inserting distorted data to influence the global model. This work presents strong anomaly detection and outlier removal methods to detect and eliminate tainted updates.

**Data reconstruction attacks** Adversaries might possibly recreate sensitive patient information by exploiting shared model updates, despite the raw data being stored on local devices.

**Membership inference attacks** Malicious entities may try to identify if certain persons provided data for training, which could lead to privacy breaches.

**Model inversion attacks** An attacker can deduce critical details about specific data points used in training by utilizing the trained model, even if the original data is not directly obtainable.

**Evasion attacks** It involves manipulating input data to deceive the intrusion detection model, allowing attackers to avoid detection.

**51% attack** The act of gaining control of more than half of a network's computational power in order to alter transactions.

**Smart contract vulnerabilities** They refer to exploitable code found within smart contracts designed for automating transactions.

### 4.2.2 Mitigation strategies

**Differential privacy** The proposed technique incorporates differential privacy techniques, adding controlled noise to model updates. This thwarts potential data reconstruction

attempts from shared information while preserving acceptable model accuracy.

**Secure aggregation** To protect individual contributions during model update aggregation, secure aggregation techniques using cryptographic methods have also been added.

**Minimal data sharing** The proposed solution focuses on using only the necessary data for intrusion detection and key exchange to reduce dangers related to sharing data, even if it is anonymized.

**Data minimization techniques** The method incorporates data minimization strategies to collect, store, and process just necessary data elements.

**Input validation** This work implements rigorous input validation to detect fraudulent alterations.

**Secure multi-party computation (SMC)** Explore Secure Multi-Party Computation (SMC) approaches for model aggregation to prohibit access to raw updates.

**Privacy-preserving blockchain platforms** This article investigates the utilization of privacy-preserving blockchain platforms such as consortium blockchains or permissioned blockchains. These restrictions limit access to approved parties, reducing the risk of unauthorized entities gaining access to information.

**Zero-knowledge proofs** Zero-Knowledge Proofs involve demonstrating the accuracy of information without disclosing specific data. This method can be utilized to show adherence to regulations without revealing confidential information on the blockchain.

**Code audit** This article utilizes thorough code audits and secure coding standards to reduce smart contract vulnerabilities.

**Transparency with stakeholders** This paper also demonstrates that ensuring transparency with stakeholders involves clear communication about data collection, processing, and security measures used in the system with healthcare institutions, patients, and regulatory agencies.

**Adherence to regulations** This strategy ensures strict compliance with healthcare data privacy requirements, such as HIPAA, which is essential to protecting patient data and adhering to ethical standards.

**Comprehensive security assessments** The proposed technique involves doing regular vulnerability assessments and penetration testing on all system components to proactively detect and address any security issues.

**Secure communication channels** Utilizing encrypted communication channels to safeguard data exchange within the system, preventing eavesdropping and data manipulation.

**Simulated attacks** This research evaluates the system's performance by testing it against different simulated attack scenarios that exploit known weaknesses in FL and ANN models.

**Metrics development** Develop metrics for quantifying the system's performance degradation under attack.

**Adaptive response** The research suggests methods for detecting attacks in real-time and taking corrective steps, such as omitting suspect model updates or using different detection methodologies.

**Threat modeling and monitoring** Thorough threat modeling is conducted to identify probable attack scenarios, followed by the implementation of continuous security monitoring mechanisms to swiftly notice and respond to security problems.

**Open communication** The proposed method ensures clear and open communication with stakeholders on the security features, limitations, and potential hazards of the system.

**Collaboration with security experts** Engaging with security researchers and specialists to regularly assess the system's security status and detect new risks in the IoMT environment.

### 4.3 IoMT security challenges

**Weak device security** There are a lot of IoMT devices that have little computing power and inadequate security safeguards, which makes them susceptible to hacking.

**Data privacy** IoMT devices collect sensitive medical data, and breaches can expose this data.

**Legacy systems** Many healthcare facilities use outdated systems that are not secure.

**Limited resources** Healthcare organizations may lack the resources to properly secure their IoMT systems.

**Long lifespans** Medical devices often have long lifecycles, making them vulnerable to new threats as security protocols become outdated.

The proposed method plays a significant role in establishing a strong security framework for IoMT by addressing several critical aspects:

**Privacy preservation** Through the use of FL, the system circumvents the need to store confidential patient data on a centralized server or transfer it over the network. Instead, the emphasis is on exchanging encrypted model updates (weights) that do not include any patient data. This greatly diminishes the likelihood of data breaches that might potentially expose confidential medical information.

**Collaborative learning** FL allows IoMT devices to gain knowledge from one another's experiences without exchanging raw data. This enables the IDS to adjust to various attack patterns throughout the network, which might result in enhanced anomaly detection capabilities.

**Reduced communication overhead** FL minimizes the need to transfer raw data to a central server, which results in a reduction in the amount of data transmission. On the other

hand, this is very necessary for Internet of Things devices that have constrained data transmission capability and limited resources.

**Enhanced intrusion detection** The ongoing process of learning in FL enables the IDS system to remain up-to-date on evolving threats and attack patterns. This improves the IoMT network's IDS as a whole.

**Stronger key exchange** In order to generate unique and complex encryption keys, the ANN-based key exchange protocol employs machine learning algorithms. These keys are employed to provide safe communication between various devices. This increases the level of difficulty for attackers attempting to decipher the encryption keys and obtain unauthorized entry to valuable data.

### 4.4 Protect against cloud security breaches

This research primarily revolves around FL and ANN-based key exchange. Its main strategy for preventing security breaches is to avoid keeping sensitive patient data on the cloud. Conventional cloud breaches often focus on centralized databases that store user information. This FL-based method provides security through:

**Limited cloud storage** A central server, maybe located in the cloud, receives only encrypted model updates, also known as weights, and not the raw data. The weights themselves do not include any patient data.

**Privacy-preserving learning** The FL process allows devices to collaboratively learn from each other without sharing raw data. This makes it much harder for attackers to steal sensitive information even if they breach the cloud server.

However, the system does have a central point of vulnerability:

**Central server compromise** If a malicious actor gains access to the central server, they could potentially manipulate the aggregated model updates before sending them back to the devices. This could degrade the effectiveness of the intrusion detection system.

Here's how the system mitigates this risk to some extent:

**Secure aggregation functions** Cryptographic techniques are used to ensure the integrity of the weight aggregation process, making it difficult for attackers to inject misleading information.

## 5 Deployment

### 5.1 Understanding adoption barriers

**Organizational resistance** This study acknowledges possible concerns like workflow interruption, resource constraints, or unfamiliarity with the technology. We will proactively interact with healthcare institutions to comprehend their unique



challenges and customize implementation tactics to reduce interruptions and optimize benefits.

**Regulatory barriers** This study has comprehensively examined all pertinent regulations concerning data privacy, security, and ethics in the designated deployment areas. The system will be created to adhere to these standards and overcome any possible regulatory obstacles.

**Cultural factors** This study acknowledges the presence of diverse cultures and norms across different healthcare systems and institutions. We will examine cultural sensitivity and customize communication tactics to ensure that the suggested system is in line with local traditions and values.

## 5.2 Strategies for gaining buy-in

**Stakeholder engagement** This work allows for engagement with healthcare professionals, administrators, patients, and lawmakers through workshops, demonstrations, and open communication channels to address their concerns and promote collaboration.

**Cost–benefit analysis** This work allows an in-depth cost-benefit analysis to illustrate the possible long-term cost reductions and enhanced security advantages in comparison to current alternatives, showcasing the system’s value for healthcare organizations.

**Pilot projects** This work allows for the carrying out of pilot projects in specific healthcare settings to showcase the system’s efficacy and resolve any potential concerns in a controlled context prior to wider implementation.

**Change management strategy** This work allows for the creation of a detailed plan to assist in the shift to the new system, offering users sufficient training, support, and communication during the transition. Transparency and Trust: This work allows for prioritizing transparency during the implementation phase to build trust. Open communication of the system’s features, constraints, and data security procedures is essential for establishing trust with all parties involved.

## 5.3 Interoperability strategies

**Device agnostic design** The system is designed to be device-agnostic, meaning it is not dependent on specific IoMT device types or communication protocols. Pre-processing data at the edge ensures compatibility with different data formats and allows for adjustment to different device setups.

**Standardized data representation** This work allows for the utilization of known data interchange protocols, such as HL7 FHIR, which can assure uniform and interoperable data formats in the FL framework. This allows for the smooth transfer of information between the system and various IoMT devices, platforms, and organizations.

**API-based communication** This work allows for the utilization of a well-defined Application Programming Interface (API) that enables integration with current healthcare information systems and IoMT platforms, fostering interoperability among various technology ecosystems.

## 5.4 Adherence to relevant standards

**Security standards** This work rigorously follows security standards like HIPAA and GDPR to guarantee data privacy and confidentiality while the system is in operation.

**Communication protocols** This system will adhere to established communication protocols such as LoRaWAN (Long Range Wide Area Network) or BLE (Bluetooth Low Energy), facilitating seamless integration with current IoMT infrastructure that employs these protocols.

## 5.5 Openness and collaboration

**Open-source components** This encourages cooperation and communal input to create standardized procedures and collections that improve compatibility throughout the IoMT field.

**Standardization initiatives** This work allows participants to participate in standardization discussions and contribute to attempts to build interoperable communication protocols and data interchange formats in the IoMT community.

## 5.6 Addressing user needs and usability

**Healthcare professionals** This work allows for the identification of specific workflows and pain points where the system could assist or potentially hinder.

**Patients** Understand privacy desires, comfort levels, and potential usability barriers depending on their technical proficiency.

**IT staff** Focus on implementation ease, alert system integration, and dashboard design for effective monitoring.

**User-centered design (UCD) principles** Throughout system development, this work adopts a user-centered design process.

**Clear alert system** Intrusion detection alerts must be understandable, providing actionable information without overwhelming technical jargon.

**Educational resources** Developing concise guides, tutorials, and FAQs to support users navigating system capabilities.

## 5.7 Feedback and iterative improvement

User feedback is crucial throughout the entire system's lifecycle. This work proposes:

**Feedback channels** Implement intuitive mechanisms for reporting issues or suggestions directly within user interfaces.

**Regular reviews** Analyze feedback data to prioritize updates and identify common usability challenges.

**Adaptability** Maintain flexibility in the system's design to incorporate adjustments driven by user experiences.

## 5.8 Practical implication of the proposed technique

**Enhanced privacy** By leveraging FL, the system avoids storing sensitive patient data on a central server, significantly reducing the risk of data breaches. This aligns well with data privacy regulations in healthcare.

**Improved intrusion detection** FL enables collaborative learning across devices, allowing the IDS to adapt to diverse attack patterns and potentially improve detection accuracy. This is crucial for protecting against constantly evolving cyber threats.

**Reduced communication overhead** FL minimizes data transfer compared to sending raw data to a central server. This is beneficial for resource-constrained IoMT devices with limited bandwidth, a common characteristic in real-world deployments.

**Stronger key exchange** The ANN-based key exchange offers dynamic and complex keys, making them more difficult to crack compared to traditional methods. This enhances communication security in the IoMT network.

**Decentralized management** Both FL and ANN-based key exchange eliminate reliance on a central server, reducing a single point of failure and potentially improving system resilience.

## 5.9 Limitations of the proposed technique

**Data availability** The quality and diversity of the training data used for the FL-IDS and ANN-based key exchange can significantly impact their performance. The study might acknowledge limitations related to the specific dataset used.

**Security analysis** While the approach aims to be secure, a thorough evaluation of its long-term security against potential attacks is likely acknowledged as an ongoing area of research.

**Integration challenges** Integrating this new security approach with existing IoMT infrastructure and communication protocols is a complex task. The

study might mention limitations related to the practical implementation and need for further development.

## 6 Results and analysis

Utilizing BoT-IoT dataset, the proposed methodology will be assessed with ANN and contrasted with XGBoost before the findings are contrasted with relevant tasks that also examined the same dataset. The method will be evaluated and then evaluated on various datasets.

### 6.1 Working environment

The proposed device simulation was conducted on a system equipped with the following specifications: an NVidia Quadro P3000 GPU with CUDA v8.0, an Intel Xeon 6th generation CPU (1 socket, 8 cores, 16 threads), and 32 GB of RAM. Additionally, the simulation utilized TensorFlow as the primary framework, along with Python ML and FL libraries such as Keras and PyTorch.

### 6.2 Data preprocessing

The dataset is preprocessed in the manner described below:

1. Deleting its columns to eliminate pointless functionality and unneeded network traffic data.
2. Inserting mean or median numbers in place of Null/NaN numbers.
3. For the purpose of understanding and evaluating the suggested model, the LabelEncoder method of the Scikit Learn package is used to convert symbolically or non-numerically labels into numerical values ranging from 0 and  $n\_classes-1$ .
4. Normal and attack traffic properties were given the Boolean categorization labels of 0 and 1, accordingly.
5. To preserve the real distribution of the characteristic, normalize high-dimensional characteristics that use the MinMaxScalar method in the scikit-learn package to a limit of (0, 1).
6. To imitate the information of 5 endpoints, divide the dataset into 5 small user datasets (acting like gateways for the monitored devices). The datasets were divided so that every user with the local model could detect unusual traffic or attacks.
7. Given that there is no overlap between the training and test flows, the analyzed dataset is randomly separated into training and testing (80%) and a test sets (20%).

### 6.3 Empirical validation strategies

**Public datasets** This study examines publicly accessible network intrusion detection datasets, specifically BoT-IoT. The underlying methodologies receive robust initial validation.

**Open-source simulators** This study explores the adaptation of current simulators, such as NS-3, to accommodate the IoMT by incorporating federated features for controlled and repeatable testing scenarios.

**Performance metrics** The study uses common intrusion detection metrics such as detection rate (recall), precision, false positive rate, accuracy, and F1-scores to offer a comprehensive assessment.

**Security evaluation** This article examines the resilience against typical adversarial attacks such as data poisoning and evasion in a federated environment.

**Comparative analysis** This article contrasts centralized ML systems with federated alternatives to provide context for the outcomes.

### 6.4 BoT-IoT dataset

#### 6.4.1 Definition of the dataset

The BoT-IoT dataset [37] is utilized for gauge how well the suggested approached performed. The Cyber Range Laboratory at the University of New South Wales (UNSW) constructed an actual test bed scenario in order to build an IoT traffic-based dataset. The Canberra Testbed was built up with fake and real-world scenarios genuine IoT traffic that is both typical and subject to threat actors, providing and over 73 million data, includes 46 characteristics in each raw and available in a 16.7 GB CSV format size. The dataset includes common occurrences as well as three different categories of threats the following terms and their additional subgroups: information collection, Denial of Service, and identity fraud to create 10 different strikes. To make the suggested model's training and testing procedures easier, a diminished set (5%)

of the initial complete The researchers of the complete dataset offered an abstracted version of data set. This 5% dataset is made up of 4 files totaling 1.07 GB and 3 million records, roughly [37].

This dataset was specifically designed for the aim of evaluating intrusion detection systems within the context of the IoT, and it is publicly available. Network traffic, both benign and malicious, collected from an IoMT simulation environment makes up the data set.

#### Benefits of using BoT-IoT

**Realistic scenarios** The dataset encompasses various attack types commonly seen in IoMT networks, allowing researchers to evaluate the system's effectiveness against real-world threats.

**Diversity** The data includes traffic from different types of IoMT devices, reflecting the heterogeneity of devices typically found in healthcare settings.

**Public availability** Using a publicly available dataset allows for reproducibility and comparison of research findings with other studies.

#### 6.4.2 Selection of features

Utilizing Eq. 14 for combined entropy and correlation coefficient downtrend, the research [37] offered the choice of the top 10 attributes from the BoT-IoT dataset ("DstIP", "seq", "SrcIP", "srate", "mean", "max", "min", "stddev", "drate" and "state number"), which helps to improve the outcome of the suggested IDS and reduces errors, as illustrated in table 3.

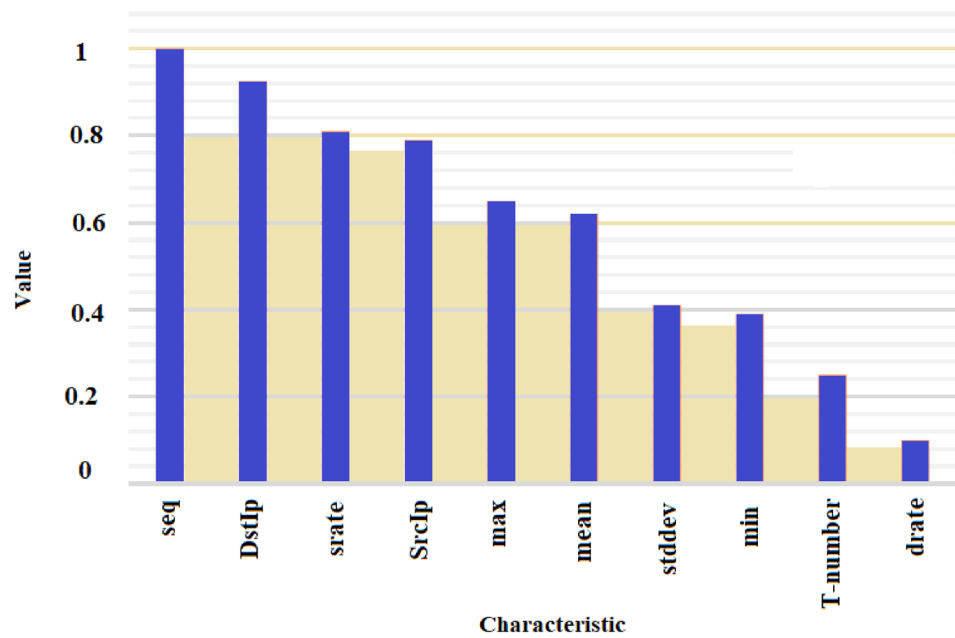
$$Entp = - \sum_y \sum_z (q(y, z) * \log q(y, z)) \quad (14)$$

Higher entropy numbers showed less information gain and show the variability of the data. First,  $p * p$  table, where  $n$  represent the number of features, was produced by applying Eq. 1 to determine the pairwise Shannon joint entropy [42]. The mean combined entropy was then calculated, and scores were normalized to establish a numerical value per

**Table 3** The major characteristics of BoT-IoT are described

| State number      | Numerial reflection of attribute state                     |
|-------------------|------------------------------------------------------------|
| Srate             | Packets per second from source to target                   |
| N_IN_Conn_P_DstIP | Number of links coming in from each target IP              |
| N_IN_Conn_P_SrcIP | The quantity of connections from one origins IP to another |
| Seq               | Sequential order for argus                                 |
| Stddev            | The aggregated records' standard deviation                 |
| Mean              | Mean record duration for all records                       |
| Max               | Maximum period for aggregated records                      |
| Min               | Minimum period of aggregated records                       |
| Drate             | Packets per second from target to source                   |

**Fig. 5** Rating features-based on information gained



|        |   | Prediction          |                     |
|--------|---|---------------------|---------------------|
|        |   | 1                   | 0                   |
| Actual | 1 | True Positive (TP)  | False Negative (FN) |
|        | 0 | False Positive (FP) | True Negative (TN)  |

**Fig. 6** Confusion matrix

features. The strength of the link between the attributes in the dataset is first measured using a matrix made to use the Pearson correlation factor. Magnitude of its correlations result,

which has a ranging between [1, 1], reveals how strongly two feature vectors are correlated. Secondly, a calculation and normalization within [0, 1] are done to determine the correlation value for each characteristic of the dataset. The desired feature was one with a high average combined entropy and a lower average correlation value.

Equation 15 was used to compute the acceptance of the top 10 BoT-IoT dataset characteristics along with Information Gain (IGA) in order to demonstrate the significance of these features. The most distinctive characteristics are ('DstIP', 'max', 'seq', 'SrcIP', and 'srate'), whereas the other attributes ('mean', 'min', 'stddev', 'drate' and 'state number') have less info gains and contribute fewer to the IDS, as depicted in Fig. 5.

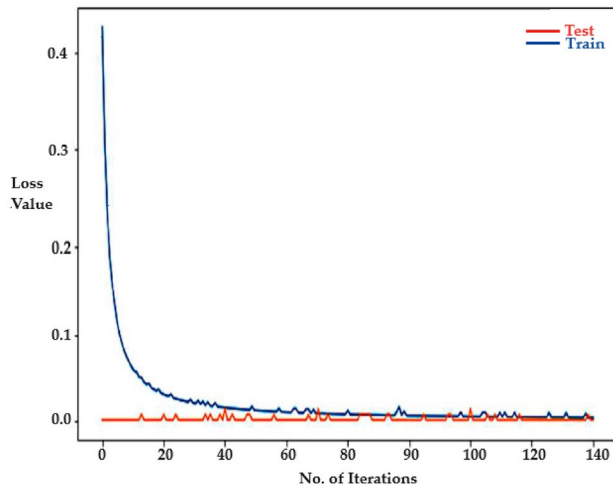
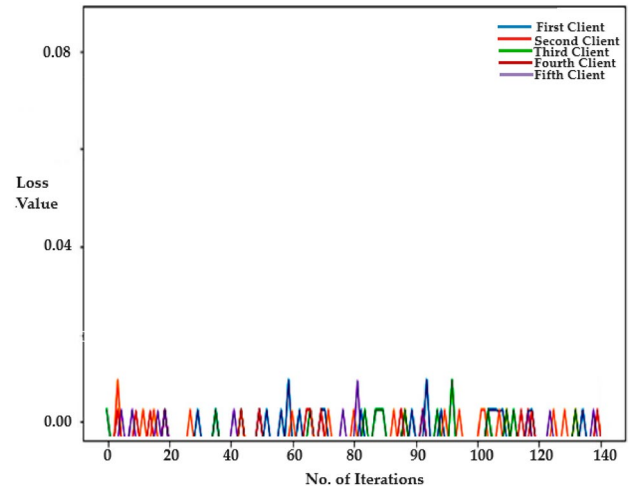
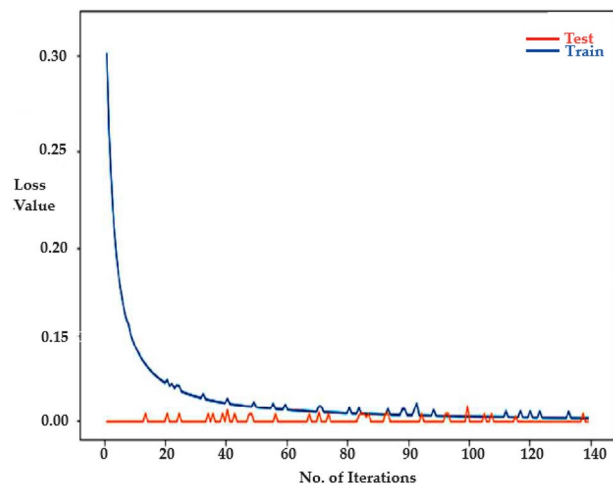
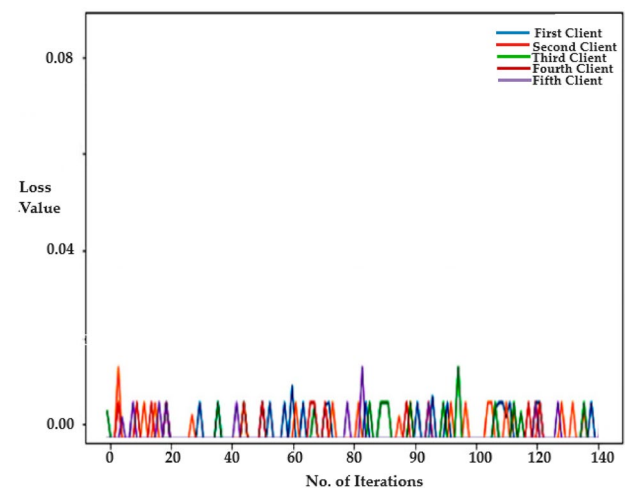
$$IGA(T, R) = F(T) - \sum_{j=1}^l Q_j F(T, R_j) \quad (15)$$

**Table 4** Principal metrics for effectiveness

| Metric               | Equation                    | Definition                                                                                                                                          |
|----------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Exactness            | $\frac{UQ}{UQ+GQ}$          | Ratio of positive events that were correctly predicted to all predictions of positive outcomes                                                      |
| Accuracy             | $\frac{UQ+UP}{UQ+UP+GQ+GP}$ | Ratio of cases with accurate forecasts to total projected occurrences                                                                               |
| Specificity          | $\frac{UP}{UP+GQ}$          | Ratio of correctly predicted negative cases to the entire category of available negative data                                                       |
| False alarm rate     | $\frac{GQ}{GP+GQ}$          | Ratio of false-positive alarms to all false-positive warnings or alerts resulting from incorrect prediction                                         |
| F1-score             | $\frac{2UQ}{2UQ+GQ+GP}$     | The hybrid measure, which is useful for classes with unbalanced data, displays the model's entire effectiveness with regard to precision and recall |
| Recall (sensitivity) | $\frac{UQ}{UQ+GP}$          | The proportion of positively predicted instances to all already available positively classified data                                                |

**Table 5** Evaluation of proposed technique's effectiveness utilizing ANN in comparison to XGBoost utilizing BoT-IoT dataset

| ML Model                       | ANN               |                   | XGBOOST           |                   |
|--------------------------------|-------------------|-------------------|-------------------|-------------------|
| Dataset version                | Complete features | Ten best features | Complete features | Ten best features |
| Exactness (level of detection) | 100%              | 100%              | 99.36%            | 99.38%            |
| Accuracy                       | 99.99%            | 99.99%            | 98.40%            | 98.96%            |
| F1-score                       | 99.99%            | 99.99%            | 99.47%            | 99.47%            |
| Selectivity                    | 88.89%            | 100%              | 56.98%            | 57.12%            |
| False alarm rate               | 11.11%            | 0%                | 43.02%            | 42.88%            |
| Recall                         | 99.99%            | 99.99%            | 99.59%            | 99.57%            |

**Fig. 7** Edge gateways (clients) of the suggested algorithm's training and testing losses averaged over BoT-IoT (complete features)**Fig. 9** Losses during algorithm testing for every client using: BoT-IoT (full features)**Fig. 8** Edge gateways (clients) of the suggested algorithm's training and testing losses averaged over: BoT-IoT. (best features)**Fig. 10** Losses during algorithm testing for every client using: BoT-IoT (best features)

## 6.5 The sensing model evaluation technique

The proposed method's identification model was assessed for exactness, predictive accuracy, accuracy, recall, F1-score, and false alarm rates in the categorization of threat detection. Following are some of the key indicators to inspect the efficiency of the model Boolean classification. The CM (Confusion Matrix) is frequently utilized to provide a fuller understanding when assessing an efficiency of the model [9], as illustrated in Fig. 6.

The primary units is frequently utilized to gauge how well a classification approach performs by assessing how well a detection approach can differentiate between various types of network traffic [5]. The measurements utilize in this investigation are displayed in Table 4.

## 6.6 Using the BoT-IoT dataset to measure the proposed algorithm

The suggested method's classification method will be assessed for Boolean classification based on its correctness, Exactness (level of detection), recall (sensitive), Selectivity, F1-score, and rate of false alarms. To replicate the data of five edge devices in these test cases, the BoT-IoT dataset [39] was split into five individual client datasets (clients). A dataset has been used to assess the detection model using "Full Features" and "Top 10 Aspects" on two ML, ANN and XGBoost.

This study will evaluate the suggested FL-based IDS and ANN-based key exchange mechanism using XGBoost and ANN models. When it comes to categorization, XGBoost is one of the most effective machine learning models out there. It is possible to compare the FL-based IDS's performance to that of a centralized dataset trained XGBoost model. This evaluates the performance of the IDS in collaborative learning on distributed data in comparison to a typical, centralized method. The outcomes are summarized in table 5. There is a decent minimum threshold for the loss, which typically happens after around 150 communications sessions. Figures 7, 8 display the average testing and training losses for edge access points (clients) using the proposed ANN approach using the BoT-IoT dataset (amazing functionality, great features respectively). Losses from testing are shown in Figs. 9 and 10 for each client separately.

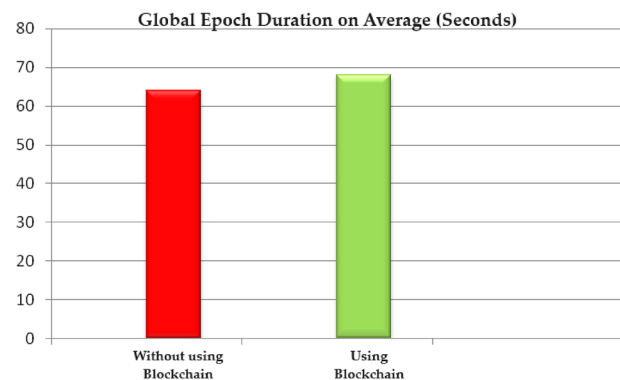
The findings demonstrate that ANN outperforms XGBoost due to greater precision in both dataset variants (99.99%), while XGBoost has lower precision for the full characteristics variant and optimal characteristics variants of the data set, accordingly, of 98.4% and 98.6%. These outcomes are generated by averaging the outcomes of all edge local models. For easy controls, ANN has

a higher detection accuracy (100%))than XGBoost (99.36% and 99.38%) in the BoT-IoT with sophisticated features and superb service, suggesting more accurate positive forecasts of attacks. There will be fewer false positives as a result of using our suggested remedy. If we compare XGBoost's recall rates for the BoT-IoT with the most recent features 99.99% and the optimum attributes 99.57%, we can see that ANN detection models cover more ground when it comes to real positive samples. When it comes to the most up-to-date and efficient BoT-IoT, ANN outperforms XGBoost (99.99% and 99.47% accordingly) in terms of F1-score.

These F1-scores highlight that ANN outperforms XGBoost particularly when considering false positives and false negatives.

Moreover, ANN outperforms XGBoost in terms of specificity, indicating wider protection of genuine negative events. Furthermore, ANN has a smaller level of false alarms than XGBoost, that means fewer erroneous warnings of the threats.

Ultimately, it is clear from the findings that the suggested method using ANN detection model outperforms using other more modern techniques as XGBoost. This is due to the significant advancements made to ANN model that make them better and much more appropriate for attack detection. Neural network possesses the subsequent features: the ability to comprehend complicated and nonlinear associations, as in IoT [76]; after studying from the connections of the preliminary information, it can generalize and forecast unknown information; It has superior efficiency when dealing with the diversity of information in IoT, for example in ICUs in medical practices, and it does not impose any constraints on the dataset and its dissemination. The proposed method is appropriate for medical practices due to its capability to predict intrusions and threats fast and correctly, simplicity, and suitability for edge-distributed systems with the constrained computing power of medical IoT.



**Fig. 11** Global epoch duration on average with and without blockchain



**Table 6** On the BoT-IoT dataset, comparability with relevant research

| Reference                | Model            | Type of classification | Accuracy (%) | Exactness (level of detection) (%) | Recall (%) | F1-score (%) | Mode        | Blockchain integration |
|--------------------------|------------------|------------------------|--------------|------------------------------------|------------|--------------|-------------|------------------------|
| Aldhaheeri et al. [2]    | DeepDCA (DCASNN) | Binary                 | 98.73        | 99.17                              | 98.36      | 98.77        | Centralized | No                     |
| Fatani et al. [20]       | CNNTSODE         | Binary                 | 99.99        | 99.99                              | 99.99      | 99.99        | Centralized | No                     |
|                          |                  | Multi                  | 99.04        | 99.04                              | 99.04      | 99.04        |             |                        |
| Ferrag et al. [22]       | DNN              | Multi                  | 98.37        |                                    |            |              | Centralized | No                     |
|                          | RNN              |                        |              |                                    |            |              |             |                        |
|                          | CNN              |                        |              |                                    |            |              |             |                        |
| Ferrag and Maglaras [21] | RNN              | Multi                  | 98.20        |                                    |            |              | Centralized | No                     |
| Kumar et al. [40]        | RF               | Multi                  | 99.99        | 99.99                              | 99.99      | 99.99        | Centralized | Yes                    |
|                          | XGBoost          |                        | 99.99        | 87.77                              | 94.36      | 87.90        |             |                        |
| Pokhrel et al. [60]      | Naive Bayes      | Binary                 | 51.5         |                                    |            |              | Centralized | No                     |
|                          | KNN              |                        | 92.1         |                                    |            |              |             |                        |
|                          | ANN              |                        | 82.8         |                                    |            |              |             |                        |
| Hussain et al. [30]      | NB               | Binary                 | 52.18        | 79.67                              | 99.70      | 69.50        | Centralized | No                     |
|                          | KNN              |                        | 99.48        | 99.65                              | 99.68      | 99.58        |             |                        |
|                          | RF               |                        | 99.51        | 99.70                              | 99.79      | 99.65        |             |                        |
|                          | Log R            |                        | 99.50        | 95.28                              | 90.39      | 94.70        |             |                        |
|                          | DT               |                        | 99.47        | 99.69                              | 99.79      | 99.63        |             |                        |
| Shafiq et al. [69]       | decision tree    | Multi                  | 99.99        | 97.10                              | 94.27      | 98.95        | Centralized | No                     |
|                          | Naive Bayes      |                        | 97.49        | 56.28                              | 57.95      | 98.44        |             |                        |
|                          | Random Forest    |                        | 99.98        | 95.05                              | 91.37      | 99.99        |             |                        |
|                          | SVM              |                        | 97.80        | 57.89                              | 43.24      | 98.48        |             |                        |
| Huong et al. [29]        | ANN              | Multi                  | 99.9         |                                    |            |              | Centralized | No                     |
|                          |                  |                        | 92.5         |                                    |            |              | Federated   |                        |
| Proposed                 | ANN              | Binary                 | 99.99        | 100                                | 99.99      | 99.99        | Federated   | No                     |

## 6.7 FL-based blockchain evaluation

Due to the high volume of weight changes, the FL process will not be practical over public blockchain that relies over evidence of effort and agreement mechanism. Just like in Hyperledger or Ethereum, the ledger's production of new blocks would be too sluggish. In light of this, the suggested

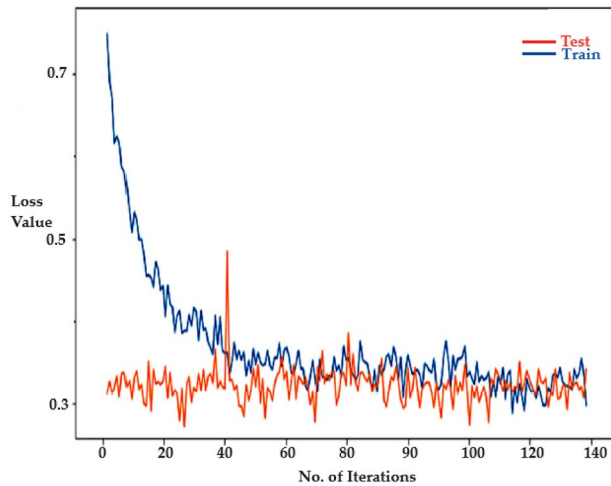
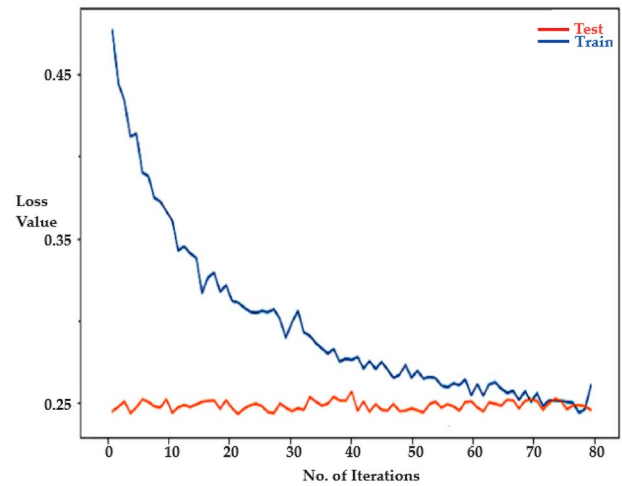
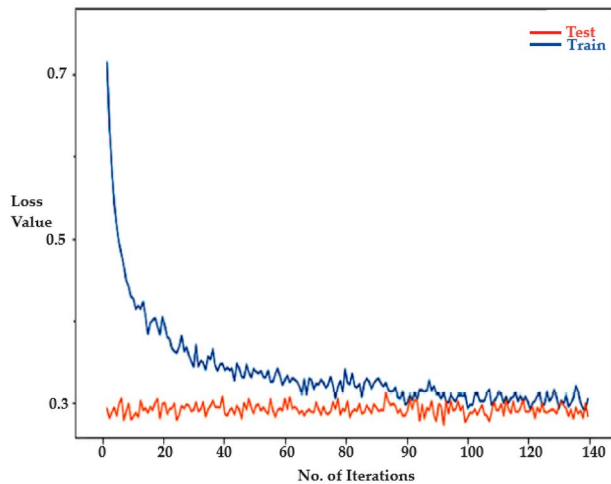
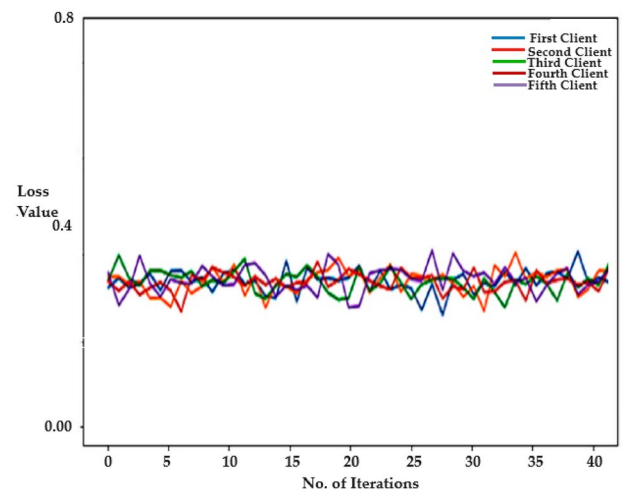
approach relies on a personal, obtaining blockchain that enables a consensus process using block identities and a programmable round robin agreement model excluding evidence of effort. The efficiency of FL is not significantly impacted by the blockchain method's increased complexity. The edge node records the weight's occurrence on the blockchain at the conclusion of each locally upgrade, and the

**Table 7** List of datasets

| Dataset                  | Description                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bot Net IoT UNB [79]     | Dataset based on internet-connected devices is split into check datasets and coaching datasets (using 7 different botnet kinds) and check datasets (using 16 different botnet kinds) and has four categories of characteristics (rely on bytes, rely on packet, period, and rely on behavior)                                                                                         |
| CSE-CIC-IDS2018 UNB [80] | A network traffic-based dataset comprising 80 network flow characteristics and 7 different botnet kinds has been suggested by the CSE (Communications Security Establishment) and the CIC (Canadian Institute for Cybersecurity)                                                                                                                                                      |
| KDD Cup 99 UCI [78]      | A dataset depends on network traffic has about 4,900,000 vectors in it. R2L (Remote-to-Local assault), U2R (User-to-Root attack), DoS (Denial-of-Service) attack and probing attack are the four categories into which the botnet kinds are split. Each category contains 41 characteristics, those are separated into 3 groups (the fundamentals, traffic, and data characteristics) |

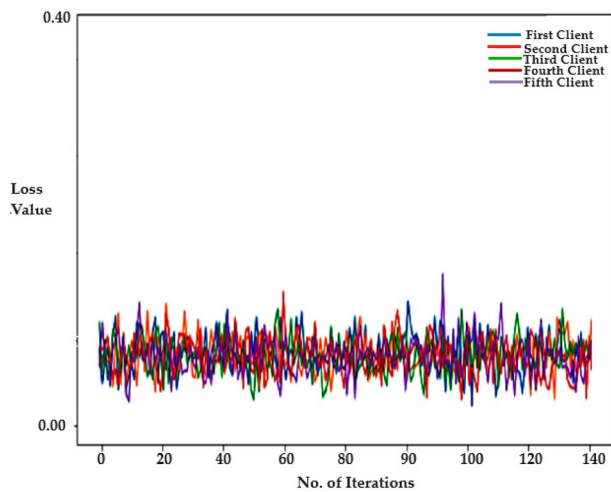
**Table 8** Outcomes of proposed method's evaluation with several datasets

| Dataset                | Exactness (level of detection) | Recall (sensitive) | F1-score | Specificity | Accuracy | False alarm rate |
|------------------------|--------------------------------|--------------------|----------|-------------|----------|------------------|
| Bot Net IoT            | 1.0000                         | 0.9753             | 0.9871   | 0.9995      | 0.9761   | 0.0003           |
| Bot-IoT (10 features)  | 1.0000                         | 0.9999             | 0.9999   | 1.0000      | 0.9999   | 0.0000           |
| Bot-IoT (all features) | 1.0000                         | 0.9999             | 0.9999   | 0.8889      | 0.9999   | 0.1111           |
| CSE-CIC-IDS2018        | 0.4461                         | 0.8581             | 0.5870   | 0.8589      | 0.8588   | 0.1411           |
| KDD Cup 99             | 0.9709                         | 0.9491             | 0.9599   | 0.9928      | 0.9840   | 0.0072           |

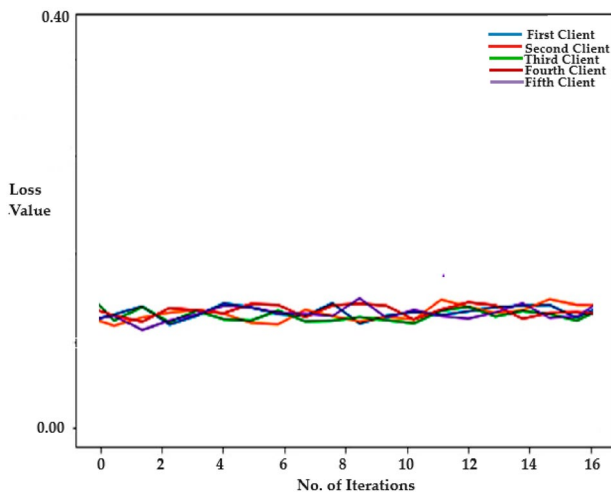
**Fig. 12** The average of edge gateways' (clients') training and testing losses for the suggested method utilizing CSE-CIC-IDS2018**Fig. 14** The average of edge gateways' (clients') training and testing losses for the suggested method utilizing KDD Cup 99**Fig. 13** The average of edge gateways' (clients') training and testing losses for the suggested method utilizing Bot Net IoT**Fig. 15** Losses in evaluating the suggested algorithm for each client utilizing the CSE-CIC-IDS2018

CPCC calculates the regional weight' mean result. To reduce latency, we recommend aligning the averaging mechanism with the duration of the block formation since the majority of blockchains generate new blocks at specified times.

Training took about 54 min on average, with approximately to 1 min greater or less spent on a single network edge, or  $54 \pm 1$ . We have observed that the blockchain method raises complexity with only a small effect on the FL time,



**Fig. 16** Losses in evaluating the suggested algorithm for each client using Bot-IoT



**Fig. 17** Losses in evaluating suggested algorithm for each client using KDD Cup 99

**Table 9** Frequency test's p value

| Procedures                | p value  |
|---------------------------|----------|
| Proposed                  | 0.702672 |
| CVTPM Dong and Huang [18] | 0.512374 |
| VVTPM Jeong et al. [32]   | 0.538632 |
| GAN-DLTPM Sarkar [66]     | 0.584326 |
| Dolecki and Kozera [17]   | 0.537924 |
| Liu et al. [52]           | 0.516357 |
| Karakaya et al. [35]      | 0.1329   |

which results in certain executing computational complexity. Experimental measurements of the effect both prior to and after blockchain implementation are possible. Both with and

without utilizing blockchain, the estimated time demands per period are determined. Using figure 11, it is projected that the overhead will be under 3% (2 s on average each epoch), while still ensuring total visibility and integrity of the decentralized training procedure.

## 6.8 Comparison with existing techniques

This section compares the suggested approach with related researches that use well-known ML and DL methods which were analyzed upon BoT-IoT dataset, according to Table 6. The proposed method can reduce false positives in contrast to the mentioned methods since its accuracy rate, remember, and F1-score are better than those of the other techniques in the majority of categories with the desired accuracy. The study [40] used blockchain technology for 99.99% accurate storage, however the learning IDS is centralized, putting patient data in danger. The paper [61], proposes a strategy to store local weights, compute averages, and supply current weights using with an auto-encoder model and blockchain technology. The accuracy reached was 97%, which is lower than the suggested technique when using the CICIDS2017 dataset for assessment. The central database is however removed from the blockchain network, so the global model weights are not updated. The research paper [29] developed a methodology that offers an IDS in FL mode using ANN, but with less accuracy 92.5% over proposed method and without using blockchain advancements for defending against poisons assaults.

## 6.9 Utilizing various datasets to test the proposed method with ANN

Here, utilizing various datasets and comparison to the BoT-IoT, the suggested method's performance measures will be assessed on the basis of correctness, remember (sensitive), F-score, selectivity, accuracy, and false alarm level of detection. These datasets are described in Table 7. To simulate the data from 5 edge devices, the datasets were divided into 5 smaller user datasets, following the exact test case structure as in BoT-IoT. The results presented in Table 8 indicate that the BoT-IoT dataset yields the most reliable and specific outcomes. Being an IoT traffic-based dataset with a diverse range of botnets, it is highly suitable for evaluating IoT applications, particularly those in medical practices. Figures 12, 13, 14, which use the CSE-CIC-IDS2018, BoT-IoT, and Knowledge discovery in databases Cup 99 datasets, accordingly, demonstrate the mean test and training losses of edge gateways of the suggested approach. Loss from testing are shown in Figs. 15, 16, 17 for every user separately.

**Table 10** Time spent synchronizing between proposed and CVTPM

| <i>I</i> value | Proposed approach's synchronization period in the anti-Hebbian (cycle) | CVTPM approach's synchronization period in the anti-Hebbian (cycle) [18] |
|----------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 5              | 378,72                                                                 | 391, 05                                                                  |
| 10             | 738,31                                                                 | 759, 72                                                                  |
| 15             | 1082,62                                                                | 1191, 18                                                                 |
| 20             | 1625,09                                                                | 1869, 96                                                                 |
| 25             | 2395,35                                                                | 2615, 82                                                                 |
| 30             | 3147,27                                                                | 3593, 48                                                                 |
| 35             | 4431,05                                                                | 4735, 62                                                                 |
| 40             | 5714,64                                                                | 5992, 08                                                                 |
| 45             | 6674,49                                                                | 8187, 51                                                                 |
| 50             | 8791,12                                                                | 9434, 26                                                                 |

**Table 11** Time spent synchronizing between proposed and VVTPM

| <i>I</i> value | Proposed approach's synchronization period in the random walk (cycle) | VVTPM approach's synchronization period in the random walk (cycle) [32] |
|----------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| 5              | 382,09                                                                | 426,98                                                                  |
| 10             | 674,18                                                                | 734,54                                                                  |
| 15             | 1005,51                                                               | 1163,24                                                                 |
| 20             | 1649,77                                                               | 1793,17                                                                 |
| 25             | 2483,49                                                               | 2585,02                                                                 |
| 30             | 3283,08                                                               | 3378,16                                                                 |
| 35             | 4118,57                                                               | 4854,23                                                                 |
| 40             | 4673,78                                                               | 5386,81                                                                 |
| 45             | 6392,02                                                               | 7389,08                                                                 |
| 50             | 7785,49                                                               | 8784,19                                                                 |

## 6.10 Performance comparisons of proposed key exchange with existing methods

An outcome of the frequency test for the synchronized neural key reveals a ratio that falls somewhere between 1 and 0. The results is 0.702672, it is superior to the output 0.584326 in [66], 0.538632 in [32], 0.512374 in [18], 0.516357 in [52], 0.1329 in [35], 0.537924 in [17]. Table 9 views the *p* value distinctions for the frequency test.

The time of the Anti-Hebbian harmonization process is compared between the suggested approach and the CVTPM technique in Table 10. A comparison of the suggested synchronization time with the synchronization time of CVTPM reveals that the recommended time is significantly shorter. In contrast to previous tactics, the Anti-Hebbian approach is significantly quicker.

Table 11 use the RWT (Random Walk Technique) to differentiate the proposed method and time coordination techniques of VVTPM. Since the proposed method uses of the Random Walk learning mechanism, it correlates considerably quicker than the VVTPM.

## 7 Conclusion and future scope

This article presents a strong security architecture for health-care's Internet of Medical Things (IoMT), which combines blockchain technology with an Intrusion Detection System (IDS) driven by Federated Learning (FL) and an ANN-based key exchange. The research started by looking at where state-of-the-art approaches fall short, and then it presented blockchain-based approaches to efficient key exchange and intrusion detection. The neural coordination technique was employed to enable secure key exchange across open networks, leveraging the rapid coordination of ANNs through optimal weights derived from collaborative input processing. Furthermore, this research introduced an edge-cloud IDS approach, which incorporates the proposed method, creating a streamlined shared ledger suitable for deployment on network edges within medical settings. These healthcare solutions harness the potential of IoMT while minimizing computational, storage, and power requirements. The integration of this method at the network edge allows for

swift identification of various botnets and quicker response to attacks, effectively preventing data from leaving the network as threats approach. Additionally, the utilization of FL models with reduced datasets significantly reduces computational demands. In tackling a major challenge of FL, which is data contamination, this approach utilizes a public blockchain to document the weights of local models within linked blocks on a public ledger. These linked blocks connect the server component in the subsequent cloud layer for aggregation, secured through cryptographic hashing algorithms and smart contracts, ensuring tamper resistance. Despite the additional complexity introduced by blockchain, the system's impact on time resources remains minimal, preserving the overall efficiency of the decentralized system. Key parameters like as recall, accuracy, F1-score, specificity, precision, and false alarm rate were examined empirically using the BoT-IoT dataset, which is based on IoT traffic. The outcomes show that the suggested approach not only reduces false alarms and improves accuracy, but also offers more accurate intrusion prevention than current ML and DL models. Both the theoretical underpinnings and the practical application of this novel strategy have been thoroughly investigated in this work. By conducting tests that mimic real-world IoMT settings, we want to prove its efficacy and set a new standard for security and intrusion detection in the dynamic IoMT environment. Improving the method's effectiveness through threat type and origin categorization, making it compatible with other hardware platforms, and streamlining data collecting techniques to overcome latency issues caused by slower participants are all potential areas for future research.

**Data availability** Data will be made available on request.

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